

Algebra

CHAPTER 6

Inequalities Involving Means

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6

Inequalities Involving Means

SIMPLE MEANS OF TWO POSITIVE NUMBERS

Let A , G and H be arithmetic, geometric and harmonic means of two positive numbers a and b .

$$\text{Then, } A = \frac{a+b}{2}, G = \sqrt{ab} \text{ and } H = \frac{2ab}{a+b}$$

These three means are related as $A \geq G \geq H$.

Proof:

We have

$$A = \frac{a+b}{2}, G = \sqrt{ab} \text{ and } H = \frac{2ab}{a+b}$$

$$\Rightarrow A - G = \frac{a+b}{2} - \sqrt{ab} = \frac{(\sqrt{a} - \sqrt{b})^2}{2} \geq 0$$

$$\Rightarrow A \geq G \quad (1)$$

$$\begin{aligned} G - H &= \sqrt{ab} - \frac{2ab}{a+b} \\ &= \sqrt{ab} \left\{ \frac{a+b-2\sqrt{ab}}{a+b} \right\} \\ &= \frac{\sqrt{ab}}{a+b} (\sqrt{a} - \sqrt{b})^2 \geq 0 \end{aligned}$$

$$\Rightarrow G \geq H \quad (2)$$

From (1) and (2), we get $A \geq G \geq H$

Clearly, $A = G = H$ if $a = b$.

ILLUSTRATION 6.1

Prove that $(ab + xy)(ax + by) > 4abxy$; where $a, b, x, y > 0$.

Sol. Using A.M. $\geq$ G.M., we have

$$\frac{ab + xy}{2} > \sqrt{abxy}$$

$$\Rightarrow ax + xy > 2\sqrt{abxy} \quad (1)$$

$$\text{Similarly, } ax + by > 2\sqrt{abxy} \quad (2)$$

Multiplying (1) and (2), we get

$$(ab + xy)(ax + by) > 4abxy$$

ILLUSTRATION 6.2

Prove that $b^2c^2 + c^2a^2 + a^2b^2 > abc(a + b + c)$, where $a, b, c > 0$.

Sol. Using A.M. $\geq$ G.M., we have

$$\frac{b^2c^2 + c^2a^2}{2} > \sqrt{(b^2c^2)(c^2a^2)}$$

$$\Rightarrow b^2c^2 + c^2a^2 > 2abc^2 \quad (1)$$

Similarly,

$$c^2a^2 + a^2b^2 > 2a^2bc \quad (2)$$

$$a^2b^2 + b^2c^2 > 2ab^2c \quad (3)$$

Adding (1), (2) and (3), we get

$$2b^2c^2 + 2c^2a^2 + 2a^2b^2 > 2abc^2 + 2bca^2 + 2cab^2$$

$$\Rightarrow b^2c^2 + c^2a^2 + a^2b^2 > abc(a + b + c)$$

ILLUSTRATION 6.3

Prove that $\frac{2}{b+c} + \frac{2}{c+a} + \frac{2}{a+b} < \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$, where $a, b, c > 0$.

Sol. Using A.M. $>$ H.M., we get

$$\frac{\frac{1}{a} + \frac{1}{b}}{2} > \frac{2}{a+b}, \frac{\frac{1}{b} + \frac{1}{c}}{2} > \frac{2}{b+c} \text{ and } \frac{\frac{1}{c} + \frac{1}{a}}{2} > \frac{2}{c+a}$$

Adding, we get

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} > \frac{2}{a+b} + \frac{2}{b+c} + \frac{2}{c+a}$$

ILLUSTRATION 6.4

If a, b , and c are distinct positive real numbers such that $a + b + c = 1$, then prove that $\frac{(1+a)(1+b)(1+c)}{(1-a)(1-b)(1-c)} > 8$.

Sol. $a + b + c = 1$

$$\text{or } a = 1 - b - c$$

$$\Rightarrow 1 + a = (1 - b) + (1 - c) > 2\sqrt{(1 - b)(1 - c)} \quad (\text{Using A.M.} \geq \text{G.M.})$$

$$\text{Similarly, } 1 + b > 2\sqrt{(1 - c)(1 - a)}$$

$$\text{and } 1 + c > 2\sqrt{(1 - a)(1 - b)}$$

Multiplying, we get

$$(1 + a)(1 + b)(1 + c) > 8(1 - a)(1 - b)(1 - c)$$

$$\text{or } \frac{(1 + a)(1 + b)(1 + c)}{(1 - a)(1 - b)(1 - c)} > 8$$

ILLUSTRATION 6.5

Find the minimum value of $4^{\sin^2 x} + 4^{\cos^2 x}$ for $x \in R$.

Sol. Using A.M. $\geq$ G.M., we have

$$\frac{4^{\sin^2 x} + 4^{\cos^2 x}}{2} \geq \sqrt{4^{\sin^2 x} \cdot 4^{\cos^2 x}}$$

$$\therefore 4^{\sin^2 x} + 4^{\cos^2 x} \geq 2\sqrt{4^{\sin^2 x + \cos^2 x}}$$

$$\Rightarrow 4^{\sin^2 x} + 4^{\cos^2 x} \geq 2\sqrt{4^1}$$

Hence, the minimum value of $4^{\sin^2 x} + 4^{\cos^2 x}$ is 4.

ILLUSTRATION 6.6

If $\log_2(a+b) + \log_2(c+d) \geq 4$. Then find the minimum value of the expression $a+b+c+d$.

Sol. $\log_2(a+b) + \log_2(c+d) \geq 4$

$$\therefore \log_2\{(a+b)(c+d)\} \geq 4$$

$$\therefore (a+b)(c+d) \geq 2^4$$

$$\text{Now } \frac{(a+b) + (c+d)}{2} \geq \sqrt{(a+b)(c+d)} \geq 2^2$$

$$\text{or } a+b+c+d \geq 8$$

Hence, the minimum value of $a+b+c+d$ is 8.

ILLUSTRATION 6.7

Find all real solutions to $2^x + x^2 = 2 - \frac{1}{2^x}$.

Sol. Given equation is $2^x + \frac{1}{2^x} = 2 - x^2$

Using A.M. $\geq$ G.M., we have $2^x + \frac{1}{2^x} \geq 2$

$$\text{But } 2 - x^2 \leq 2.$$

$$\text{So, given equation holds only if } 2^x + \frac{1}{2^x} = 2 - x^2 = 2$$

Therefore, $x = 0$ is the only solution.

ILLUSTRATION 6.8

Find all positive real solutions to

$$4x + \frac{18}{y} = 14, 2y + \frac{9}{z} = 15, 9z + \frac{16}{x} = 17.$$

Sol. Adding all the given equations, we have

$$4x + \frac{16}{x} + 2y + \frac{18}{y} + 9z + \frac{9}{z} = 46 \quad (1)$$

Now, using A.M. $\geq$ G.M., we have

$$4x + \frac{16}{x} \geq 16, 2y + \frac{18}{y} \geq 12 \text{ and } 9z + \frac{9}{z} \geq 18.$$

$$\therefore 4x + \frac{16}{x} + 2y + \frac{18}{y} + 9z + \frac{9}{z} \geq 46$$

So, equation (1) holds only if $4x + \frac{16}{x} = 16$, $2y + \frac{18}{y} = 12$ and $9z + \frac{9}{z} = 18$.

$$\therefore 4x = \frac{16}{x}, 2y = \frac{18}{y} \text{ and } 9z = \frac{9}{z}$$

$$\therefore x = 2, y = 3 \text{ and } z = 1$$

CONCEPT APPLICATION EXERCISE 6.1

1. Prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{1}{\sqrt{bc}} + \frac{1}{\sqrt{ca}} + \frac{1}{\sqrt{ab}}$, where $a, b, c > 0$.
2. If a, b, c are real numbers such that $0 < a < 1$, $0 < b < 1$, $0 < c < 1$, $a+b+c=2$, then prove that $\frac{a}{1-a} \cdot \frac{b}{1-b} \cdot \frac{c}{1-c} \geq 8$.
3. If x, y are positive real numbers and m, n are positive integers, then prove that $\frac{x^n y^m}{(1+x^{2n})(1+y^{2m})} \leq \frac{1}{4}$.
4. If $a_i > 0$ ($i = 1, 2, 3, \dots, n$), prove that
$$\sum_{1 \leq i < j \leq n} \sqrt{a_i a_j} \leq \frac{n-1}{2} (a_1 + a_2 + \dots + a_n)$$
5. Find the minimum value of $2^{\sin x} + 2^{\cos x}$.
6. If $\log_{10}(x^3 + y^3) - \log_{10}(x^2 + y^2 - xy) \leq 2$, where x, y are positive real number, then find the maximum value of xy .
7. Prove that the greatest value of xy is $c^3/\sqrt{2ab}$, if $a^2x^4 + b^2y^4 = c^6$.
8. If $x, y \in R^+$ such that $x+y=8$, then find the minimum value of $\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right)$.

ANSWERS

5. $2^{1-\frac{1}{\sqrt{2}}}$ 6. 2500 8. $\frac{25}{16}$

SIMPLE MEANS OF n POSITIVE NUMBERS

Consider n positive quantities $a_1, a_2, a_3, \dots, a_n$.

A.M., G.M. and H.M. of these quantities are defined as

$$\text{A.M.} = \frac{a_1 + a_2 + \dots + a_n}{n}$$

$$\text{G.M.} = (a_1 \cdot a_2 \cdot \dots \cdot a_n)^{1/n}$$

$$\text{H.M.} = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}$$

ILLUSTRATION 6.9

Let A , G and H be arithmetic, geometric and harmonic means, respectively, between three given positive numbers a , b and c . Then prove that the equation having a , b and c as its roots is

$$x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0.$$

Sol. We have

$$A = \frac{a+b+c}{3} \Rightarrow a+b+c = 3A$$

$$G = (abc)^{1/3} \Rightarrow abc = G^3$$

$$\frac{1}{H} = \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} = \frac{ab+bc+ca}{3abc}$$

$$\Rightarrow \frac{3G^3}{H} = ab+bc+ca$$

The equation having a , b and c as its roots is

$$x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc = 0$$

$$\text{or } x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$$

ILLUSTRATION 6.10

Let a_1, a_2 and $a_3, \dots, a_n$ be positive real numbers in geometric progression. For each n , let A_n, G_n and H_n be, respectively, arithmetic mean, geometric mean, and harmonic mean of $a_1, a_2, \dots, a_n$. Find an expression for the geometric mean of $G_1, G_2, \dots, G_n$ in terms of $A_1, A_2, \dots, A_n, H_1, H_2, \dots, H_n$.

Sol. $a_1, a_2, a_3, \dots$ are in G.P.

Let r be the common ratio.

For k such quantities,

$$\begin{aligned} G_k &= (a_1 a_2 \dots a_k)^{1/k} \\ \Rightarrow G_k &= a_1 (r^{1+2+\dots+(k-1)})^{1/k} \\ \Rightarrow G_k &= a_1 r^{\frac{k-1}{2}} \end{aligned} \quad (1)$$

$$\begin{aligned} A_k &= \frac{a_1 + a_2 + \dots + a_k}{k} \\ \Rightarrow A_k &= \frac{a_1 (1 + r + \dots + r^{k-1})}{k} \end{aligned}$$

$$\Rightarrow A_k = \frac{a_1 (r^k - 1)}{(r - 1) \cdot k} \quad (2)$$

$$\begin{aligned} H_k &= \frac{k}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}} \\ \Rightarrow H_k &= \frac{a_1 k}{1 + \frac{1}{r} + \dots + \frac{1}{r^{k-1}}} = \frac{a_1 k}{1 - \frac{1}{r^k}} \cdot \frac{1}{1 - \frac{1}{r}} \end{aligned}$$

$$\Rightarrow H_k = \frac{a_1 k (r - 1) r^{k-1}}{r^k - 1} \quad (3)$$

From (1), (2) and (3), we get

$$G_k = (A_k H_k)^{1/2}$$

$$\Rightarrow \prod_{k=1}^n G_k = \prod_{k=1}^n (A_k H_k)^{\frac{1}{2}}$$

$$\Rightarrow \left(\prod_{k=1}^n G_k \right)^{1/n} = (A_1 A_2 \dots A_n \times H_1 H_2 \dots H_n)^{1/2n}$$

AM-GM INEQUALITY

Like A.M.-G.M. inequality for two positive real numbers we have A.M.-G.M. inequality for n real numbers also.

For n positive real numbers $a_1, a_2, a_3, \dots, a_n$, we have

$$\frac{\sum_{i=1}^n a_i}{n} \geq \sqrt[n]{\prod_{i=1}^n a_i}, \text{ where } \prod_{i=1}^n a_i = a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_n$$

Proof:

We have several methods to prove this inequality. One of the methods is using principle of mathematical induction.

$$\text{Let } P(n) \equiv \frac{\sum_{i=1}^n a_i}{n} \geq \sqrt[n]{\prod_{i=1}^n a_i} \quad (1)$$

We know that $P(2)$ is true. We have already proved A.M.-G.M. inequality for two positive real numbers.

Let us prove that $P(2n)$ is also true assuming that $P(n)$ is true.

Consider positive real numbers $a_1, a_2, \dots, a_{2n}$.

$$\begin{aligned} \text{Now, } \frac{1}{2n} \sum_{i=1}^{2n} a_i &= \frac{1}{2n} \left(\sum_{i=1}^n a_i + \sum_{i=n+1}^{2n} a_i \right) \\ &= \frac{1}{2} \left(\frac{1}{n} \sum_{i=1}^n a_i + \frac{1}{n} \sum_{i=n+1}^{2n} a_i \right) \\ &\geq \frac{1}{2} \left(\sqrt[n]{\prod_{i=1}^n a_i} + \sqrt[n]{\prod_{i=n+1}^{2n} a_i} \right) \\ &\quad \text{(Using A.M.} \geq \text{G.M. for } n \text{ values)} \\ &\geq \sqrt[n]{\prod_{i=1}^n a_i} \times \sqrt[n]{\prod_{i=n+1}^{2n} a_i} \\ &\quad \text{(Using A.M.} \geq \text{G.M. for 2 values)} \\ &= \sqrt[2n]{\prod_{i=1}^{2n} a_i} \end{aligned}$$

Thus, $P(2n)$ is true if $P(n)$ is true.

Now, let us prove that $P(n-1)$ is also true.

To show this, we take n positive real numbers $a_1, a_2, \dots, a_{n-1}$ and

$$a_n = \frac{1}{n-1} \sum_{i=1}^{n-1} a_i.$$

$$\text{Now, } \frac{1}{n} \sum_{i=1}^n a_i = \frac{1}{n} \left(\sum_{i=1}^{n-1} a_i + \frac{1}{n-1} \sum_{i=1}^{n-1} a_i \right) = \frac{1}{n-1} \sum_{i=1}^{n-1} a_i.$$

So, we have

$$\begin{aligned}
 \frac{1}{n-1} \sum_{i=1}^{n-1} a_i &= \frac{1}{n} \sum_{i=1}^n a_i \\
 &\geq \sqrt[n]{\prod_{i=1}^n a_i} \\
 &= \sqrt[n]{\left(\prod_{i=1}^{n-1} a_i\right) \left(\frac{1}{n-1} \sum_{i=1}^{n-1} a_i\right)} \\
 \Rightarrow \left(\frac{1}{n-1} \sum_{i=1}^{n-1} a_i\right)^n &\geq \left(\frac{1}{n-1} \sum_{i=1}^{n-1} a_i\right) \left(\prod_{i=1}^{n-1} a_i\right) \\
 \Rightarrow \left(\frac{1}{n-1} \sum_{i=1}^{n-1} a_i\right)^{n-1} &\geq \prod_{i=1}^{n-1} a_i \\
 \Rightarrow \frac{1}{n-1} \sum_{i=1}^{n-1} a_i &\geq \left(\prod_{i=1}^{n-1} a_i\right)^{\frac{1}{n-1}}
 \end{aligned}$$

Thus, $P(n-1)$ is also true.

So, $P(n) \Rightarrow P(2n)$ and $P(n) \Rightarrow P(n-1)$.

$\therefore P(2n) \Rightarrow P(2n-1) \Rightarrow P(2n-2)$ and so on up to $P(n+1)$ are true.

Therefore, using forward-backward induction we have proved that statement $P(n)$ is true.

Alternative Method:

We have already proved A.M.-G.M. inequality for n positive real numbers $a_1, a_2, a_3, \dots, a_n$. Let us see another proof.

Consider the function $f(x) = \log x \forall x > 0$.

We observe that $f''(x) = \frac{-1}{x^2} < 0$. Therefore, graph of function $f(x)$ is concave downward.

$$\text{So, we have } f\left(\frac{\sum_{i=1}^n a_i}{n}\right) \geq \frac{\sum_{i=1}^n f(a_i)}{n}$$

$$\Rightarrow \log\left(\frac{\sum_{i=1}^n a_i}{n}\right) \geq \frac{\sum_{i=1}^n \log(a_i)}{n}$$

$$\begin{aligned}
 \text{Now, } \frac{\sum_{i=1}^n \log(a_i)}{n} &= \frac{\log a_1 + \log a_2 + \dots + \log a_n}{n} \\
 &= \frac{\log(a_1 a_2 a_3 \dots a_n)}{n} \\
 &= \log\left((a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}\right)
 \end{aligned}$$

$$\text{So, } \log\left(\frac{\sum_{i=1}^n a_i}{n}\right) \geq \log\left((a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}\right)$$

$$\text{Therefore, } \frac{\sum_{i=1}^n a_i}{n} \geq (a_1 a_2 a_3 \dots a_n)^{\frac{1}{n}}$$

GM-HM INEQUALITY

For n positive real numbers $a_1, a_2, a_3, \dots, a_n$, we have

$$\sqrt[n]{a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_n} \geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}$$

This can be rearranged as

$$\frac{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}{n} \geq \sqrt[n]{\frac{1}{a_1} \cdot \frac{1}{a_2} \cdot \frac{1}{a_3} \cdot \dots \cdot \frac{1}{a_n}}$$

This is AM-GM inequality for positive real numbers

$$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}.$$

So, GM-HM inequality holds for n positive real numbers.

ILLUSTRATION 6.11

If $a + b + c = 1$, then prove that

$$\frac{8}{27abc} > \left\{\frac{1}{a}-1\right\}\left\{\frac{1}{b}-1\right\}\left\{\frac{1}{c}-1\right\} > 8; \text{ where } a, b, c > 0.$$

Sol. On multiplying both sides by abc , we have to prove that $8/27 > (1-a)(1-b)(1-c) > 8abc$. Now,

$$\begin{aligned}
 \frac{(1-a) + (1-b) + (1-c)}{3} &> [(1-a)(1-b)(1-c)]^{1/3} \\
 \Rightarrow \frac{3 - (a+b+c)}{3} &> [(1-a)(1-b)(1-c)]^{1/3} \\
 \Rightarrow \left(\frac{2}{3}\right)^3 &> (1-a)(1-b)(1-c) \quad (\because a+b+c=1)
 \end{aligned}$$

Also, $(a+b)/2 > (ab)^{1/2}$, $(c+b)/2 > (cb)^{1/2}$, and $(a+c)/2 > (ac)^{1/2}$. Multiplying these three inequalities, we have

$$\begin{aligned}
 \frac{a+b}{2} \cdot \frac{b+c}{2} \cdot \frac{a+c}{2} &> abc \\
 \text{or } (a+b)(b+c)(c+a) &> 8abc \\
 \text{or } (1-a)(1-b)(1-c) &> 8abc
 \end{aligned}$$

ILLUSTRATION 6.12

If $yz + zx + xy = 12$, where x, y, z are positive values, find the greatest value of xyz .

Sol. Using A.M. $\geq$ G.M., we have

$$\frac{xy + yz + zx}{3} \geq (x^2 y^2 z^2)^{1/3}$$

$$\Rightarrow \frac{12}{3} \geq (x^2 y^2 z^2)^{1/3}$$

$$\Rightarrow (xyz)_{\max} = \left(\frac{12}{3}\right)^{3/2}$$

ILLUSTRATION 6.13

If a, b and c are positive then prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Sol. We have

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

$$\Rightarrow \frac{a}{b+c} + 1 + \frac{b}{c+a} + 1 + \frac{c}{a+b} + 1 \geq \frac{3}{2} + 3$$

$$\Rightarrow \frac{a+b+c}{b+c} + \frac{a+b+c}{c+a} + \frac{a+b+c}{a+b} \geq \frac{9}{2}$$

$$\Rightarrow \frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \geq \frac{9}{2(a+b+c)}$$

Now, using A.M. $\geq$ H.M., we have

$$\frac{\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}}{3} \geq \frac{3}{(a+b) + (b+c) + (c+a)}$$

$$\Rightarrow \frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \geq \frac{9}{2(a+b+c)}$$

ILLUSTRATION 6.14

Prove that $2^n > 1 + n \sqrt{2^{n-1}}$, $\forall n > 2$ where n is a positive integer.

Sol. $2^n > 1 + n \sqrt{2^{n-1}}$

$$\Rightarrow \frac{2^n - 1}{2 - 1} > n \times 2^{(n-1)/2}$$

Now, $(2^n - 1)/(2 - 1)$ is the sum of a G.P. whose first term is 1 and common ratio is 2.

We have to prove that $1 + 2 + 2^2 + 2^3 + \dots + 2^{n-1} > n \times 2^{(n-1)/2}$.

Now,

Using A.M. $\geq$ G.M., we get

$$\frac{1+2+2^2+\dots+2^{n-1}}{n} > (1 \times 2 \times 2^2 \times 2^3 \dots 2^{n-1})^{1/n}$$

$$\text{Now, R.H.S.} = (2^{1+2+3+\dots+n-1})^{1/n}$$

$$= [2^{(n-1)n/2}]^{1/n} = 2^{(n-1)/2}$$

$$\text{Hence, } 1 + 2 + 2^2 + \dots + 2^{n-1} > n \times 2^{(n-1)/2}$$

ILLUSTRATION 6.15

If $S = a_1 + a_2 + a_3 + \dots + a_n$, $a_i \in R^+$ for $i = 1$ to n , then prove that

$$\frac{S}{S-a_1} + \frac{S}{S-a_2} + \dots + \frac{S}{S-a_n} \geq \frac{n^2}{n-1}, \forall n \geq 2$$

Sol. We have

A.M. $\geq$ H.M.

$$\Rightarrow \frac{\frac{S}{S-a_1} + \frac{S}{S-a_2} + \dots + \frac{S}{S-a_n}}{n} \geq \frac{n}{\frac{S-a_1}{S} + \frac{S-a_2}{S} + \dots + \frac{S-a_n}{S}}$$

$$\text{or } \frac{S}{S-a_1} + \frac{S}{S-a_2} + \dots + \frac{S}{S-a_n} \geq \frac{n^2 S}{nS - (a_1 + a_2 + \dots + a_n)}$$

$$\text{or } \frac{S}{S-a_1} + \frac{S}{S-a_2} + \dots + \frac{S}{S-a_n} \geq \frac{n^2 S}{nS - S}$$

$$\text{or } \frac{S}{S-a_1} + \frac{S}{S-a_2} + \dots + \frac{S}{S-a_n} \geq \frac{n^2}{n-1}$$

ILLUSTRATION 6.16

If $a_1 + a_2 + a_3 + a_4 + \dots + a_n = 1$, $\forall a_i > 0$, $i = 1, 2, 3, \dots, n$, then find the maximum value of $a_1^2 a_2 a_3 a_4 \dots a_n$.

Sol. Using A.M. $\geq$ G.M.

$$\frac{\frac{a_1}{2} + \frac{a_1}{2} + a_2 + a_3 + a_4 + \dots + a_n}{n+1}$$

$$\geq \left(\left(\frac{a_1}{2} \right)^2 \cdot a_2 \cdot a_3 \cdot a_4 \cdot \dots \cdot a_n \right)^{\frac{1}{n+1}}$$

$$\Rightarrow \left(\frac{1}{n+1} \right)^{n+1} \geq \frac{a_1^2 a_2 a_3 a_4 \dots a_n}{4}$$

$$\Rightarrow \frac{4}{(n+1)^{n+1}} \geq a_1^2 a_2 a_3 a_4 \dots a_n$$

Hence, required maximum value is $\frac{4}{(n+1)^{n+1}}$.

ILLUSTRATION 6.17

If a, b, c are positive real numbers. Then prove that $(a+1)^7 (b+1)^7 (c+1)^7 > 7^7 a^4 b^4 c^4$.

Sol. Given that a, b, c are positive real numbers. To prove that

$$(a+1)^7 (b+1)^7 (c+1)^7 > 7^7 a^4 b^4 c^4.$$

$$\text{L.H.S.} = (1+a)^7 (1+b)^7 (1+c)^7$$

$$= [(1+a)(1+b)(1+c)]^7$$

$$= [1+a+b+c+ab+bc+ca+abc]^7 \quad (1)$$

Now, A.M. $\geq$ G.M.

$$\Rightarrow \frac{a+b+c+ab+bc+ca+abc}{7} \geq (a^4 b^4 c^4)^{1/7}$$

$$\text{or } (a+b+c+ab+bc+ca+abc)^7 \geq 7^7 (a^4 b^4 c^4) \quad (2)$$

From Eqs. (1) and (2), we get

$$[(1+a)(1+b)(1+c)]^7 > 7^7 a^4 b^4 c^4$$

ILLUSTRATION 6.18

Prove that $\frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha} \geq 8$, if each term in the expression is well defined.

Sol. Let $\tan^2 \alpha = x$ and $\tan^2 \beta = y$

$$\begin{aligned} \therefore \frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha} &= \frac{(x+1)^2}{y} + \frac{(y+1)^2}{x} \\ &= \frac{x^2}{y} + \frac{y^2}{x} + \frac{1}{x} + \frac{1}{y} + 2\frac{x}{y} + 2\frac{y}{x} \end{aligned}$$

Now, using A.M. $\geq$ G.M., we get

$$\begin{aligned} \frac{\frac{x^2}{y} + \frac{y^2}{x} + \frac{1}{x} + \frac{1}{y} + \frac{x}{y} + \frac{x}{y} + \frac{y}{x} + \frac{y}{x}}{8} \\ \geq \left(\frac{x^2}{y} \cdot \frac{y^2}{x} \cdot \frac{1}{x} \cdot \frac{1}{y} \cdot \frac{x}{y} \cdot \frac{x}{y} \cdot \frac{y}{x} \cdot \frac{y}{x} \right)^{\frac{1}{8}} \\ \Rightarrow \frac{x^2}{y} + \frac{y^2}{x} + \frac{1}{x} + \frac{1}{y} + \frac{x}{y} + \frac{x}{y} + \frac{y}{x} + \frac{y}{x} \geq 8 \\ \therefore \frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha} \geq 8 \end{aligned}$$

CONCEPT APPLICATION EXERCISE 6.2

1. Prove that $\left(\frac{n+1}{2}\right)^n > n!$
2. If $a_1, a_2, \dots, a_n > 0$, then prove that $\frac{a_1}{a_2} + \frac{a_2}{a_3} + \frac{a_3}{a_4} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} > n$
3. In $\triangle ABC$, prove that $\tan A + \tan B + \tan C \geq 3\sqrt{3}$, where A, B, C are acute angles.
4. If $n \geq 1$ is a positive integer, then prove that $3^n \geq 2^n + n \cdot 6^{\frac{n-1}{2}}$.
5. If $abcd = 1$, where a, b, c and d are positive real numbers, then find the minimum value of $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$.
6. If $x, y, z > 0$ and $x + y + z = 1$, then prove that $\frac{2x}{1-x} + \frac{2y}{1-y} + \frac{2z}{1-z} \geq 3$.

ANSWERS

5. 10

WEIGHTED MEANS

If $a_i > 0$; $i = 1, 2, \dots, n$ and $w_i > 0$; $i = 1, 2, \dots, n$ then weighted means are defined as follows:

Weighted arithmetic mean, $A_w = \frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n}$

Weighted geometric mean, $G_w = (a_1^{w_1} a_2^{w_2} \dots a_n^{w_n})^{\frac{1}{w_1 + w_2 + \dots + w_n}}$

Weighted harmonic mean, $H_w = \frac{w_1 + w_2 + \dots + w_n}{\frac{w_1}{a_1} + \frac{w_2}{a_2} + \dots + \frac{w_n}{a_n}}$

For these means also, we have $A_w \geq G_w \geq H_w$

If $w_1 = w_2 = w_3 = \dots = w_n$, then these means are same as corresponding simple means.

ILLUSTRATION 6.19

Prove that

$$\left[\frac{x^2 + y^2 + z^2}{x + y + z} \right]^{x+y+z} > x^x y^y z^z > \left[\frac{x + y + z}{3} \right]^{x+y+z} \quad (x, y, z > 0)$$

$$\begin{aligned} \text{Sol. } \frac{x^2 + y^2 + z^2}{x + y + z} &= \frac{(x + x + x + \dots x \text{ times}) + (y + y + y + \dots y \text{ times}) + (z + z + z + \dots z \text{ times})}{x + y + z} \\ &> [(x \ x \dots x \text{ factors}) (y \ y \dots y \text{ factors}) (z \ z \dots z \text{ factors})]^{1/(x+y+z)} \end{aligned}$$

$$\text{or } \frac{x^2 + y^2 + z^2}{x + y + z} > (x^x y^y z^z)^{1/(x+y+z)}$$

$$\text{or } \left(\frac{x^2 + y^2 + z^2}{x + y + z} \right)^{x+y+z} > x^x y^y z^z \quad (1)$$

Now,

$$\begin{aligned} \frac{\left(\frac{1}{x} + \frac{1}{x} + \dots x \text{ times} \right) + \left(\frac{1}{y} + \frac{1}{y} + \dots y \text{ times} \right) + \left(\frac{1}{z} + \frac{1}{z} + \dots z \text{ times} \right)}{x + y + z} \\ > \left[\left(\frac{1}{x} \frac{1}{x} \dots x \text{ factors} \right) \left(\frac{1}{y} \frac{1}{y} \dots y \text{ factors} \right) \right. \\ &\quad \left. \times \left(\frac{1}{z} \frac{1}{z} \dots z \text{ factors} \right) \right]^{1/(x+y+z)} \end{aligned}$$

$$\text{or } \frac{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}{x + y + z} > \left[\frac{1}{x^x y^y z^z} \right]^{1/(x+y+z)}$$

$$\text{or } \left(\frac{3}{x + y + z} \right)^{x+y+z} > \frac{1}{x^x y^y z^z}$$

$$\text{or } \left(\frac{x + y + z}{3} \right)^{x+y+z} < x^x y^y z^z$$

$$\text{or } x^x y^y z^z > \left(\frac{x + y + z}{3} \right)^{x+y+z} \quad (2)$$

From Eqs. (1) and (2), we get

$$\left[\frac{x^2 + y^2 + z^2}{x + y + z} \right]^{x+y+z} > x^x y^y z^z > \left[\frac{x + y + z}{3} \right]^{x+y+z}$$

ILLUSTRATION 6.20

Prove that $1^1 \times 2^2 \times 3^3 \times \dots \times n^n < \left(\frac{2n+1}{3} \right)^{\frac{n(n+1)}{2}}$.

Sol. $1^2 = 1$

$$2^2 = 2 + 2$$

$$3^2 = 3 + 3 + 3$$

$\therefore$ Using weighted means

$$\frac{1 + (2+2) + (3+3+3) + \dots + (n+n+\dots n \text{ times})}{1+2+3+\dots+n} \geq (1^1 \cdot 2^2 \cdot \dots \cdot n^n)^{\frac{1}{1+2+3+\dots+n}}$$

$$\Rightarrow \frac{1+2^2+3^2+\dots+n^2}{\frac{n(n+1)}{2}} \geq (1^1 \cdot 2^2 \cdot \dots \cdot n^n)^{\frac{2}{n(n+1)}}$$

$$\Rightarrow \frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)}{2}} \geq (1^1 \cdot 2^2 \cdot \dots \cdot n^n)^{\frac{2}{n(n+1)}}$$

$$\Rightarrow \frac{2n+1}{3} \geq (1^1 \cdot 2^2 \cdot \dots \cdot n^n)^{\frac{2}{n(n+1)}}$$

$$\Rightarrow 1^1 \cdot 2^2 \cdot 3^3 \cdot \dots \cdot n^n \leq \left(\frac{2n+1}{3} \right)^{\frac{n(n+1)}{2}}$$

ILLUSTRATION 6.21

Find the greatest value of $x^2 y^3$ where x and y lie in the first quadrant on the line $3x + 4y = 5$.

Sol. Given that $3x + 4y = 5$ (1)

Since we have expression $x^2 y^3$, we consider

$$2\left(\frac{3x}{2}\right) + 3\left(\frac{4y}{3}\right) = 3x + 4y = 5$$

Using A.M. $\geq$ G.M. for weighted means, we get

$$\frac{2\left(\frac{3x}{2}\right) + 3\left(\frac{4y}{3}\right)}{2+3} \geq \left[\left(\frac{3x}{2}\right)^2 \left(\frac{4y}{3}\right)^3 \right]^{\frac{1}{5}}$$

$$\Rightarrow \left(\frac{3x+4y}{5} \right)^5 \geq \left(\frac{3}{2} \right)^2 \left(\frac{4}{3} \right)^3 x^2 y^3$$

$$\Rightarrow x^2 y^3 \leq \left(\frac{2}{3} \right)^2 \left(\frac{3}{4} \right)^3$$

$$\Rightarrow x^2 y^3 \leq \frac{3}{16}$$

So, the greatest value of $x^2 y^3$ is $\frac{3}{16}$.

ILLUSTRATION 6.22

Find the maximum value of $(7-x)^4 (2+x)^5$ when x lies between -2 and 7 .

Sol. Since $-2 < x < 7$

$$\therefore x+2 > 0 \text{ and } 7-x > 0$$

We have to find maximum value of $(7-x)^4 (2+x)^5$ or $p^4 q^5$ where $p+q=9$.

$$\begin{aligned} \frac{4\left(\frac{p}{4}\right) + 5\left(\frac{q}{5}\right)}{4+5} &\geq \left[\left(\frac{p}{4}\right)^4 \left(\frac{q}{5}\right)^5 \right]^{\frac{1}{9}} \\ \Rightarrow \left[\left(\frac{p}{4}\right)^4 \left(\frac{q}{5}\right)^5 \right]^{\frac{1}{9}} &\leq \frac{p+q}{9} \\ \Rightarrow \left(\frac{p}{4}\right)^4 \left(\frac{q}{5}\right)^5 &\leq 1 \\ \Rightarrow p^4 q^5 &\leq 4^4 5^5 \end{aligned}$$

Therefore, maximum value of $(7-x)^4 (2+x)^5$ is $4^4 5^5$.

ILLUSTRATION 6.23

Find the maximum value of xyz when $\frac{x}{1} + \frac{y^2}{4} + \frac{z^3}{27} = 1$; where $x, y, z > 0$.

Sol. We have $\frac{x}{1} + \frac{y^2}{4} + \frac{z^3}{27} = 1$

Using weighted means, we have

$$\begin{aligned} \frac{x + \frac{1}{2} \cdot \frac{y^2}{2} + \frac{1}{3} \cdot \frac{z^3}{9}}{1 + \frac{1}{2} + \frac{1}{3}} &\geq \left[x \left(\frac{y^2}{2}\right)^{1/2} \left(\frac{z^3}{9}\right)^{1/3} \right]^{\frac{1}{1+\frac{1}{2}+\frac{1}{3}}} \\ \Rightarrow \frac{1}{\frac{11}{6}} &\geq \left[x \left(\frac{y^2}{2}\right)^{1/2} \left(\frac{z^3}{9}\right)^{1/3} \right]^{\frac{6}{11}} \\ \Rightarrow xyz \left(\frac{1}{2}\right)^{1/2} \left(\frac{1}{9}\right)^{1/3} &\leq \left(\frac{6}{11}\right)^{\frac{11}{6}} \\ \Rightarrow xyz &\leq \left(\frac{6}{11}\right)^{\frac{11}{6}} \cdot \sqrt[2]{2} \cdot \sqrt[3]{9} \end{aligned}$$

So, maximum value of xyz is $\left(\frac{6}{11}\right)^{\frac{11}{6}} \cdot \sqrt[2]{2} \cdot \sqrt[3]{9}$.

CONCEPT APPLICATION EXERCISE 6.3

1. Prove that

$$\left[\frac{a^2 + b^2}{a + b} \right]^{a+b} > a^a b^b > \left\{ \frac{a+b}{2} \right\}^{a+b}; \text{ where } a, b > 0.$$

2. Prove that $a^p b^q > \left(\frac{ap + bq}{p+q} \right)^{p+q}$; where $a, b, p, q > 0$.3. Prove that $px^{q-r} + qx^{r-p} + rx^{p-q} > p + q + r$, where p, q, r are distinct numbers and $x > 0, x \neq 1$.4. Given are positive rational numbers a, b, c such that $a + b + c = 1$, then prove that $a^a b^b c^c + a^b b^c c^a + a^c b^a c^b \leq 1$.5. If a and b are positive numbers such that $a^2 + b^2 = 4$, then find the maximum value of $a^2 b$.6. Find the greatest value of $x^2 y^3 z^4$ if $x^2 + y^2 + z^2 = 1$, where x, y, z are positive.

ANSWERS

5. $\frac{16}{3\sqrt{3}}$ 6. $2^5 \times 3^{-15/2}$

MEANS OF m th POWER

Consider n positive quantities $a_1, a_2, a_3, \dots, a_n$. Then arithmetic mean of m th power is $\frac{a_1^m + a_2^m + \dots + a_n^m}{n}$, where m is any real number except zero.

We have the following inequalities for this mean:

(i) $\frac{a_1^m + a_2^m + \dots + a_n^m}{n} > \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m$; If $m < 0$ or $m > 1$

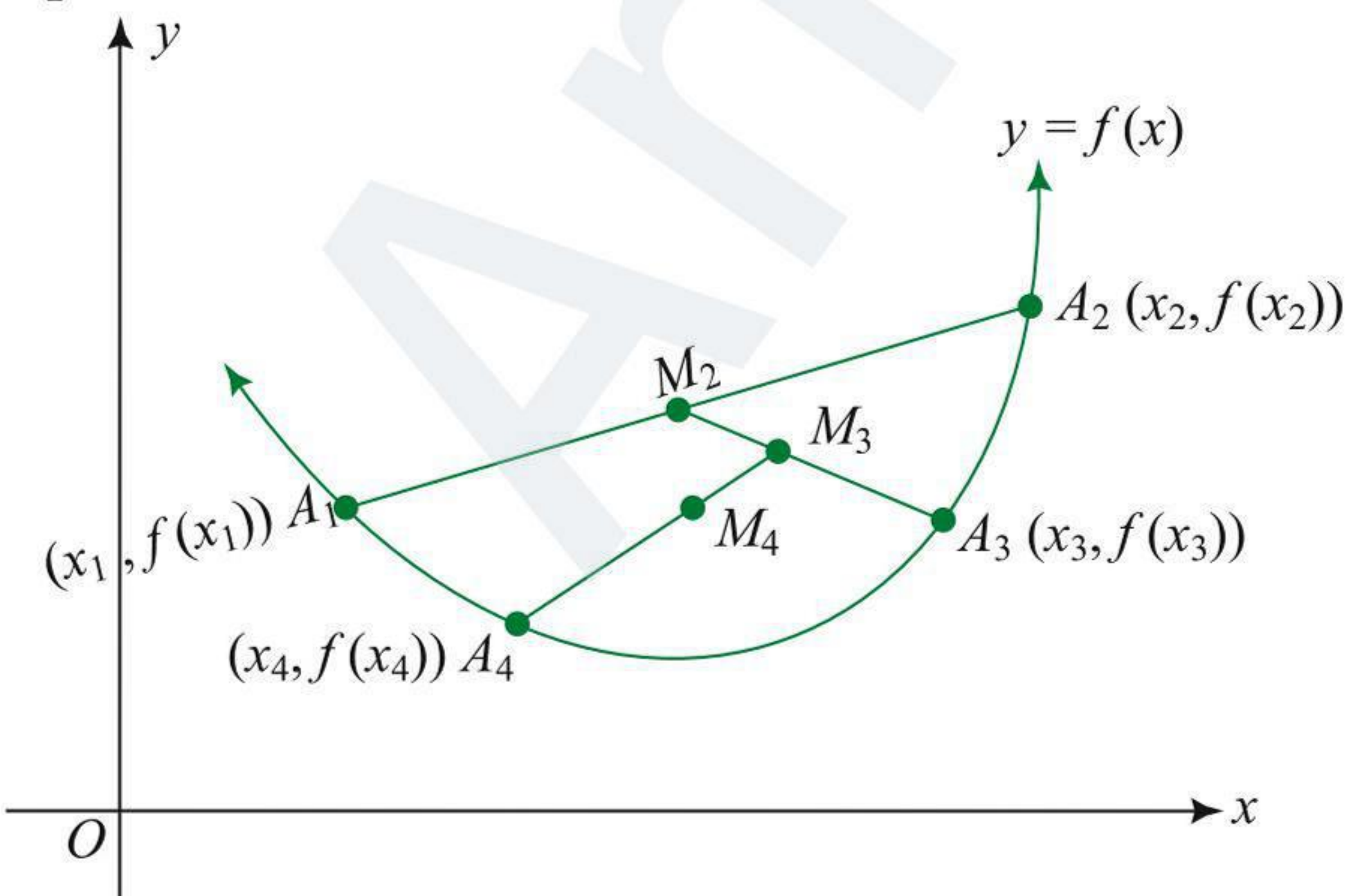
(ii) $\frac{a_1^m + a_2^m + \dots + a_n^m}{n} < \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m$; If $0 < m < 1$

(iii) $\frac{a_1^m + a_2^m + \dots + a_n^m}{n} = \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m$; If $m = 0, 1$ or

$$a_1 = a_2 = a_3 \dots = a_n$$

Proof:

(i) Consider the following graph of $y = f(x)$, which is concave upward.



Consider points $A_1(x_1, f(x_1))$ and $A_2(x_2, f(x_2))$ on the graph.

Midpoint of A_1A_2 is $M_2 \left(\frac{x_1 + x_2}{2}, \frac{f(x_1) + f(x_2)}{2} \right)$.

Now, join M_2 and $A_3(x_3, f(x_3))$.

Let point M_3 on M_2A_3 be such that $M_2M_3 : M_3A_3 = 1 : 2$.

So, coordinates of M_3 are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{f(x_1) + f(x_2) + f(x_3)}{3} \right)$.

Further on M_3A_4 , consider point M_4 such that $M_3M_4 : M_4A_4 = 1 : 3$

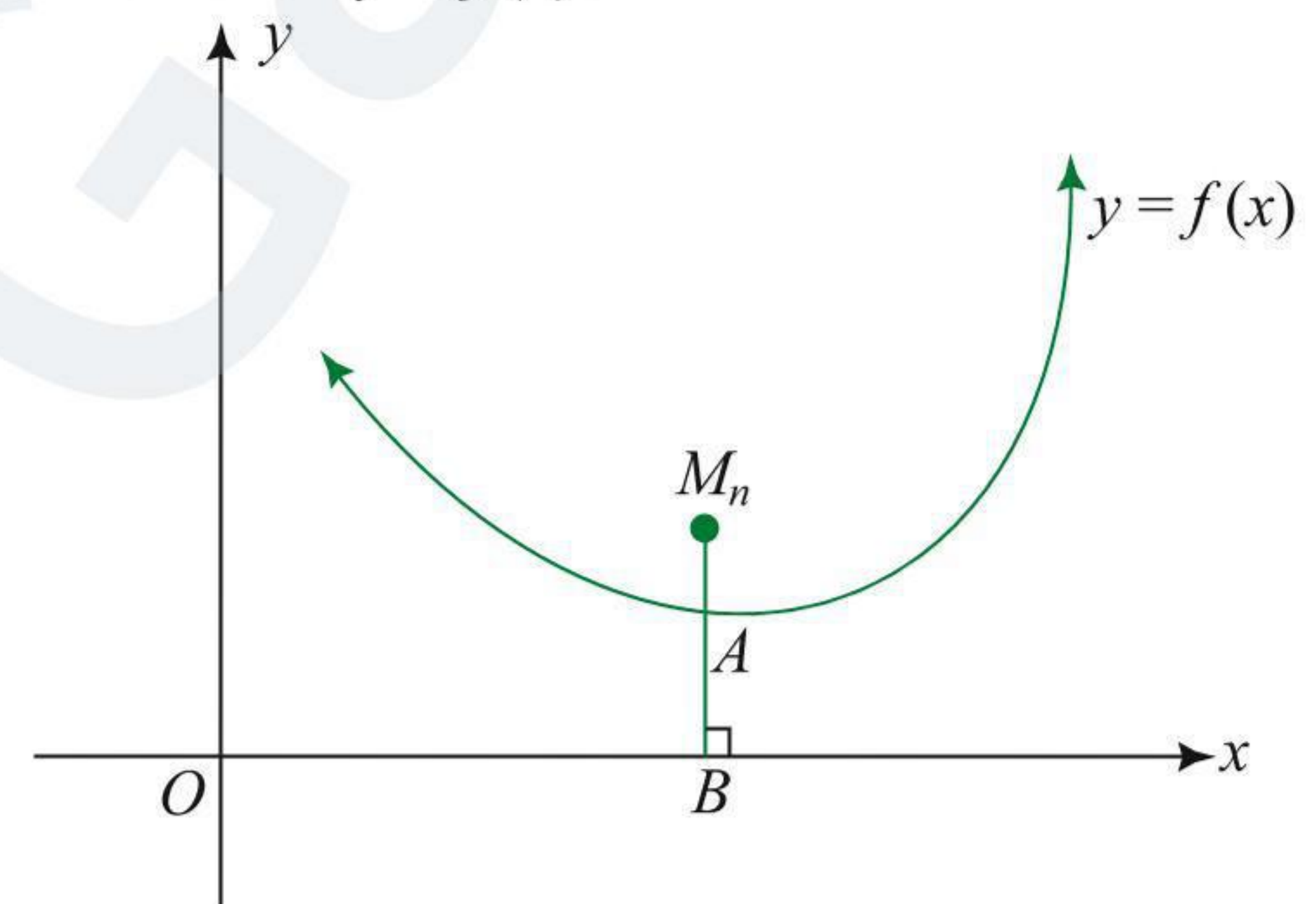
So, coordinates of M_4 are

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{f(x_1) + f(x_2) + f(x_3) + f(x_4)}{4} \right).$$

Similarly, we get point M_n whose coordinates are

$$\left(\frac{\sum_{i=1}^n x_i}{n}, \frac{\sum_{i=1}^n f(x_i)}{n} \right).$$

All the points $M_2, M_3, \dots, M_n$ lie in the region which is above the graph of the function $y = f(x)$.



In the above figure, from point M_n , draw vertical line, which intersects the graph of $y = f(x)$ at A and x -axis at B .

Coordinates of point A are $\left(\frac{\sum_{i=1}^n x_i}{n}, f \left(\frac{\sum_{i=1}^n x_i}{n} \right) \right)$.

Clearly, $M_n B > AB$

$$\therefore \frac{\sum_{i=1}^n f(x_i)}{n} > f \left(\frac{\sum_{i=1}^n x_i}{n} \right).$$

Thus, for concave upward graph, we have above inequality.

Now, graph of the function is concave upward in the interval where second derivative of the function is positive.

For $y = f(x) = x^m$, where $x > 0$ and $m < 0$ or $m > 1$,

$$\frac{dy}{dx} = mx^{m-1} \text{ and } \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Since $x > 0$ and $m < 0$ or $m > 1$, $\frac{d^2y}{dx^2} > 0$.

So, graph of function is concave upward.

Therefore, $\frac{\sum_{i=1}^n f(x_i)}{n} > f \left(\frac{\sum_{i=1}^n x_i}{n} \right)$

$$\text{or } \frac{a_1^m + a_2^m + \dots + a_n^m}{n} > \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m$$

When the graph of the function is concave downward, point M_n lies in the region which is below the graph of the function $y=f(x)$.

In that case,

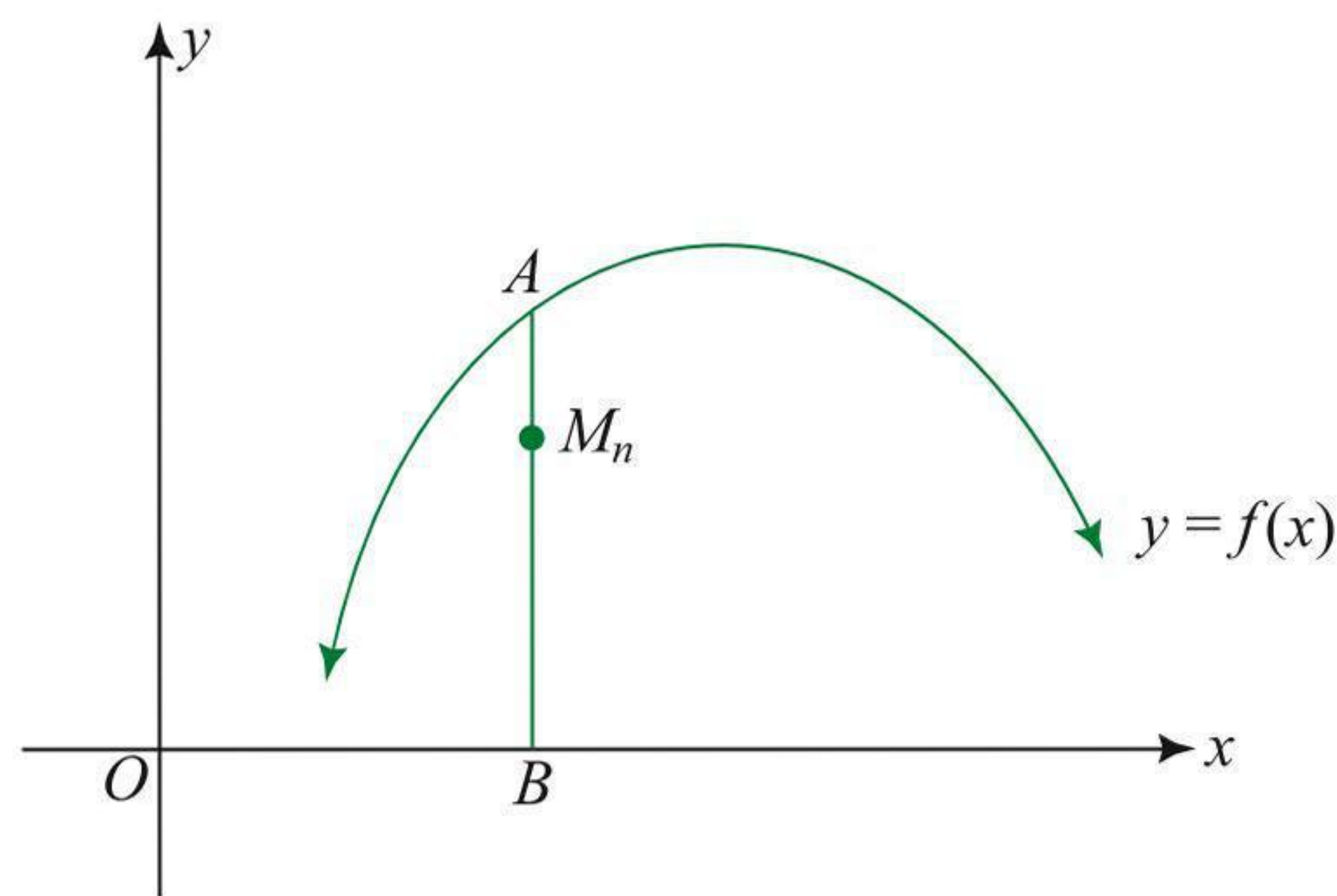
$$\frac{\sum_{i=1}^n f(x_i)}{n} < f\left(\frac{\sum_{i=1}^n x_i}{n}\right)$$

Graph of the function is downward in the interval where second derivative of the function is negative.

For $y=f(x)=x^m$, where $x>0$ and $0<m<1$,

$$\frac{d^2y}{dx^2} = m(m-1)x^{m-2} < 0$$

So, graph is concave downward.



Clearly, $M_n B < AB$

$$\text{Therefore, } \frac{a_1^m + a_2^m + \dots + a_n^m}{n} < \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m$$

ILLUSTRATION 6.24

If $a, b > 0$ such that $a^3 + b^3 = 2$, then show that $a + b \leq 2$.

Sol. We have

$$\frac{(a^3)^{1/3} + (b^3)^{1/3}}{2} \leq \left(\frac{a^3 + b^3}{2} \right)^{1/3}$$

$$\Rightarrow \frac{a+b}{2} \leq 1$$

$$\Rightarrow a+b \leq 2$$

ILLUSTRATION 6.25

If $m > 1$, $n \in \mathbb{N}$, then show that $1^m + 2^m + 2^{2m} + 2^{3m} + \dots + 2^{nm-m} > n^{1-m}(2^n - 1)^m$.

Sol. Since $m > 1$, A.M. of m th power $>$ m th power of A.M.

$$\frac{1^m + 2^m + 4^m + 8^m + \dots + (2^{n-1})^m}{n} > \left(\frac{1+2+4+\dots+2^{n-1}}{n} \right)^m$$

$$\Rightarrow 1^m + 2^m + 4^m + \dots + 2^{(n-1)m} > n \left(\frac{2^n - 1}{n} \right)^m = n^{1-m}(2^n - 1)^m$$

ILLUSTRATION 6.26

Prove that in acute angled triangle ABC , $\sec A + \sec B + \sec C \geq 6$.

Sol. Triangle ABC is acute angled.

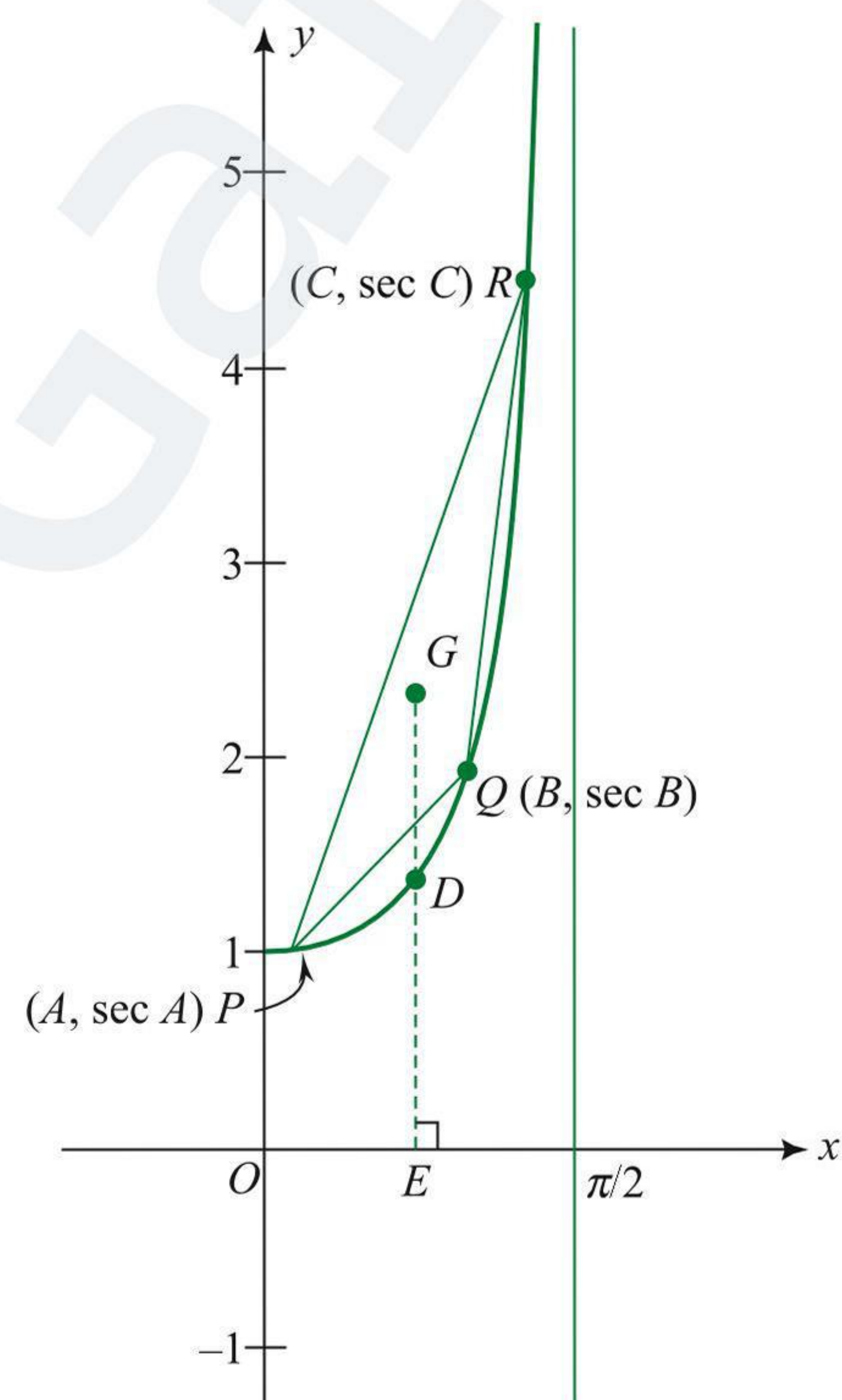
$$\therefore 0 < A, B, C < \pi/2$$

So, consider the graph of function $y = \sec x$ for $x \in (0, \pi/2)$.

The graph of function is concave upward.

Now, consider three points on the graph $P(A, \sec A)$, $Q(B, \sec B)$ and $R(C, \sec C)$.

Join these points to get triangle PQR .



Now, centroid of the triangle PQR is

$$G\left(\frac{A+B+C}{3}, \frac{\sec A + \sec B + \sec C}{3}\right).$$

Through point G , draw a line parallel to y -axis meeting graph at point D and x -axis at point E .

The abscissa of point D is $\frac{A+B+C}{3}$.

So, coordinates of point D are $\left(\frac{A+B+C}{3}, \sec\left(\frac{A+B+C}{3}\right)\right)$.

Now, clearly $GE \geq DE$

$$\therefore \frac{\sec A + \sec B + \sec C}{3} \geq \sec\left(\frac{A+B+C}{3}\right)$$

$$\Rightarrow \frac{\sec A + \sec B + \sec C}{3} \geq \sec \frac{\pi}{3}$$

$$\Rightarrow \sec A + \sec B + \sec C \geq 6$$

ILLUSTRATION 6.27

Prove that $\frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} + \frac{a^2+b^2}{a+b} > a+b+c$; where $a, b, c > 0$.

Sol. $\frac{b^2+c^2}{2} > \left(\frac{b+c}{2}\right)^2$ or $\frac{b^2+c^2}{b+c} > \frac{1}{2}(b+c)$

Similarly,

$$\frac{a^2+b^2}{a+b} > \frac{1}{2}(a+b) \text{ and } \frac{c^2+a^2}{c+a} > \frac{1}{2}(c+a)$$

Adding, we get

$$\frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} + \frac{a^2+b^2}{a+b} > a+b+c$$

ILLUSTRATION 6.28

Prove that $\frac{a^8+b^8+c^8}{a^3b^3c^3} > \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

Sol. We have to prove that

$$a^8+b^8+c^8 > a^2b^2c^2(bc+ca+ab).$$

Now,

$$\frac{a^8+b^8+c^8}{3} > \left(\frac{a+b+c}{3}\right)^8$$

or $\frac{a^8+b^8+c^8}{3} > \left(\frac{a+b+c}{3}\right)^6 \left(\frac{a+b+c}{3}\right)^2$

$$> \left[(abc)^{1/3}\right]^6 \left[\frac{a^2+b^2+c^2+2ab+2bc+2ca}{9}\right]$$

($\because$ A.M. $>$ G.M.)

But $a^2+b^2+c^2 > ab+bc+ca$

So, $\frac{a^8+b^8+c^8}{3} > a^2b^2c^2 \frac{(3ab+3bc+3ca)}{9}$

or $a^8+b^8+c^8 > a^2b^2c^2(ab+bc+ca)$

or $\frac{a^8+b^8+c^8}{a^3b^3c^3} > \frac{ab+bc+ca}{abc}$

or $\frac{a^8+b^8+c^8}{a^3b^3c^3} > \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$

ILLUSTRATION 6.29

If each of a, b and c is positive and $a+b+c=6$, then show that

$$\left(a+\frac{1}{b}\right)^2 + \left(b+\frac{1}{c}\right)^2 + \left(c+\frac{1}{a}\right)^2 \geq \frac{75}{4}.$$

Sol. Since A.M. $\geq$ H.M., we have

$$\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \frac{3}{a+b+c} = \frac{3}{6} = \frac{1}{2}$$

So, $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{3}{2}$

Now,
$$\frac{\left(a+\frac{1}{b}\right)^2 + \left(b+\frac{1}{c}\right)^2 + \left(c+\frac{1}{a}\right)^2}{3} \geq \left[\frac{\left(a+\frac{1}{b}\right) + \left(b+\frac{1}{c}\right) + \left(c+\frac{1}{a}\right)}{3}\right]^2 \geq \left[\frac{6+\frac{3}{2}}{3}\right]^2 = \frac{25}{4}$$

$\therefore \left(a+\frac{1}{b}\right)^2 + \left(b+\frac{1}{c}\right)^2 + \left(c+\frac{1}{a}\right)^2 \geq \frac{75}{4}$

CONCEPT APPLICATION EXERCISE 6.4

1. Prove that $a^4+b^4+c^4 > abc(a+b+c)$, where $a, b, c > 0$.

2. If $C_r = \frac{n!}{r!(n-r)!}$, then prove that

$$\sqrt{C_1} + \sqrt{C_2} + \cdots + \sqrt{C_n} < \sqrt{n(2^n-1)}.$$

3. If $a+b=1, a>0, b>0$, prove that $\left(a+\frac{1}{a}\right)^2 + \left(b+\frac{1}{b}\right)^2 \geq \frac{25}{2}$.

4. In ΔABC , prove that $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.

Exercises

Single Correct Answer Type

- The minimum value of $\frac{x^4 + y^4 + z^4}{xyz}$ for positive real numbers x, y, z is
 (1) $\sqrt{2}$ (2) $2\sqrt{2}$
 (3) $4\sqrt{2}$ (4) $8\sqrt{2}$
- A rod of fixed length k slides along the coordinate axes. If it meets the axes at $A(a, 0)$ and $B(0, b)$, then the minimum value of $\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2$ is
 (1) 0 (2) 8
 (3) $k^2 - 4 + \frac{4}{k^2}$ (4) $k^2 + 4 + \frac{4}{k^2}$
- The least value of $6 \tan^2 \phi + 54 \cot^2 \phi + 18$ is
 (I) 54 when A.M. $\geq$ G.M. is applicable for $6 \tan^2 \phi$, $54 \cot^2 \phi$, 18
 (II) 54 when A.M. $\geq$ G.M. is applicable for $6 \tan^2 \phi$, $54 \cot^2 \phi$, and 18 is added further
 (III) 78 when $\tan^2 \phi = \cot^2 \phi$
 (1) (I) is correct, (II) is false
 (2) (I) and (II) are correct
 (3) (III) is correct
 (4) none of the above are correct
- If $ab^2c^3, a^2b^3c^4, a^3b^4c^5$ are in A.P. ($a, b, c > 0$), then the minimum value of $a + b + c$ is
 (1) 1 (2) 3
 (3) 5 (4) 9
- If $y = 3^{x-1} + 3^{-x-1}$, then the least value of y is
 (1) 2 (2) 6
 (3) $2/3$ (4) $3/2$
- Minimum value of $(b+c)/a + (c+a)/b + (a+b)/c$ (for real positive numbers a, b, c) is
 (1) 1 (2) 2
 (3) 4 (4) 6
- If the product of n positive numbers is n^n , then their sum is
 (1) a positive integer (2) divisible by n
 (3) equal to $n + 1/n$ (4) never less than n^2
- The minimum value of $P = bcx + cay + abz$, when $xyz = abc$, is
 (1) $3abc$ (2) $6abc$
 (3) abc (4) $4abc$
- If l, m, n are the three positive roots of the equation $x^3 - ax^2 + bx - 48 = 0$, then the minimum value of $(1/l) + (2/m) + (3/n)$ equals
 (1) 1 (2) 2
 (3) $3/2$ (4) $5/2$
- If positive numbers a, b, c are in H.P., then equation $x^2 - kx + 2b^{101} - a^{101} - c^{101} = 0$ ($k \in R$) has
 (1) both roots positive
 (2) both roots negative
 (3) one positive and one negative root
 (4) both roots imaginary
- For $x^2 - (a+3)|x| + 4 = 0$ to have real solutions, the range of a is
 (1) $(-\infty, -7] \cup [1, \infty)$ (2) $(-3, \infty)$
 (3) $(-\infty, -7]$ (4) $[1, \infty)$
- If a, b, c are the sides of a triangle, then the minimum value of $\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c}$ is equal to
 (1) 3 (2) 6
 (3) 9 (4) 12
- If $a, b, c, d \in R^+ - \{1\}$, then the minimum value of $\log_d a + \log_b d + \log_a c + \log_c b$ is
 (1) 4 (2) 2
 (3) 1 (4) none of these
- If $a, b, c \in R^+$, then $\frac{bc}{b+c} + \frac{ac}{a+c} + \frac{ab}{a+b}$ is always
 (1) $\leq \frac{1}{2}(a+b+c)$ (2) $\geq \frac{1}{3}\sqrt{abc}$
 (3) $\leq \frac{1}{3}(a+b+c)$ (4) $\geq \frac{1}{2}\sqrt{abc}$
- If $a, b, c \in R^+$, then $(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$ is always
 (1) ≥ 12 (2) ≥ 9
 (3) ≤ 12 (4) none of these
- If $a, b, c \in R^+$, then the minimum value of $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2)$ is equal to
 (1) abc (2) $2abc$
 (3) $3abc$ (4) $6abc$
- If $a, b, c, d \in R^+$ and a, b, c, d are in H.P., then
 (1) $a + d > b + c$ (2) $a + b > c + d$
 (3) $a + c > b + d$ (4) none of these
- If $a, b, c \in R^+$ such that $a + b + c = 18$, then the maximum value of $a^2 b^3 c^4$ is equal to
 (1) $2^{18} \times 3^2$ (2) $2^{18} \times 3^3$
 (3) $2^{19} \times 3^2$ (4) $2^{19} \times 3^3$

19. $f(x) = \frac{(x-2)(x-1)}{(x-3)}, \forall x > 3$. The minimum value of $f(x)$ is equal to
- (1) $3 + 2\sqrt{2}$ (2) $3 + 2\sqrt{3}$
 (3) $3\sqrt{2} + 2$ (4) $3\sqrt{2} - 2$
20. If $a > 0$, then least value of $(a^3 + a^2 + a + 1)^2$ is
- (1) $64a^2$ (2) $16a^4$
 (3) $16a^3$ (4) none of these

Multiple Correct Answers Type

1. If A is the area and $2s$ the sum of the sides of a triangle, then

- (1) $A \leq \frac{s^2}{4}$ (2) $A \leq \frac{s^2}{3\sqrt{3}}$
 (3) $A < \frac{s^2}{\sqrt{3}}$ (4) none of these

2. If x, y, z are positive numbers in A.P., then

- (1) $y^2 \geq xz$ (2) $xy + yz \geq 2xz$
 (3) $\frac{x+y}{2y-x} + \frac{y+z}{2y-z} \geq 4$ (4) none of these

3. For positive real numbers a, b, c such that $a + b + c = p$, which one holds?

- (1) $(p-a)(p-b)(p-c) \leq \frac{8}{27}p^3$
 (2) $(p-a)(p-b)(p-c) \geq 8abc$
 (3) $\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \leq p$
 (4) none of these

4. If first and $(2n-1)^{\text{th}}$ terms of an A.P., G.P., and H.P. are equal and their n^{th} terms are a, b, c , respectively, then

- (1) $a = b = c$ (2) $a + c = b$
 (3) $a > b > c$ (4) $ac - b^2 = 0$

5. If $a > 0, b > 0, c > 0$ and $2a + b + 3c = 1$, then

- (1) $a^4b^2c^2$ is greatest then $a = \frac{1}{4}$
 (2) $a^4b^2c^2$ is greatest then $b = \frac{1}{4}$
 (3) $a^4b^2c^2$ is greatest then $c = \frac{1}{12}$
 (4) greatest value of $a^4b^2c^2$ is $\frac{1}{9 \cdot 4^8}$

Linked Comprehension Type

For Problems 1–3

If roots of the equation $f(x) = x^6 - 12x^5 + bx^4 + cx^3 + dx^2 + ex + 64 = 0$ are positive, then

1. which has the greatest absolute value?
 (1) b (2) c
 (3) d (4) e
2. which has the least absolute value?
 (1) b (2) c
 (3) d (4) e
3. remainder when $f(x)$ is divided by $x - 1$ is
 (1) 2 (2) 1
 (3) 3 (4) 10

For Problems 4–6

Equation $x^4 + ax^3 + bx^2 + cx + 1 = 0$ has real roots (a, b, c are non-negative).

4. Minimum non-negative real value of a is
 (1) 10 (2) 9
 (3) 6 (4) 4
5. Minimum non-negative real value of b is
 (1) 12 (2) 15
 (3) 6 (4) 10
6. Minimum non-negative real value of c is
 (1) 10 (2) 9
 (3) 6 (4) 4

Numerical Value Type

1. For $x \geq 0$, the smallest value of the function

$$f(x) = \frac{4x^2 + 8x + 13}{6(1+x)}, \text{ is } \underline{\hspace{2cm}}.$$

2. Let $x^2 - 3x + p = 0$ has two positive roots a and b , then minimum value of $\left(\frac{4}{a} + \frac{1}{b}\right)$ is $\underline{\hspace{2cm}}$.

3. If x, y , and z are positive real numbers and $x = \frac{12 - yz}{y + z}$. The maximum value of (xyz) equals $\underline{\hspace{2cm}}$.

4. If a, b , and c are positive and $9a + 3b + c = 90$, then the maximum value of $(\log a + \log b + \log c)$ is (base of the logarithm is 10) $\underline{\hspace{2cm}}$.

5. Given that x, y, z are positive reals such that $xyz = 32$. The minimum value of $x^2 + 4xy + 4y^2 + 2z^2$ is $\underline{\hspace{2cm}}$.

6. If $x, y \in R^+$ satisfying $x + y = 3$, then the maximum value of x^2y is $\underline{\hspace{2cm}}$.

7. For any $x, y \in R, xy > 0$. Then the minimum value of $\frac{2x}{y^3} + \frac{x^3y}{3} + \frac{4y^2}{9x^4}$ is $\underline{\hspace{2cm}}$.

8. Let a, b, c, d and e be positive real numbers such that $a + b + c + d + e = 15$ and $ab^2c^3d^4e^5 = (120)^3 \times 50$. Then the value of $a^2 + b^2 + c^2 + d^2 + e^2$ is $\underline{\hspace{2cm}}$.

9. Consider the system of equations $x_1 + x_2^2 + x_3^3 + x_4^4 + x_5^5 = 5$ and $x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 15$ where x_1, x_2, x_3, x_4, x_5 are positive real numbers. Then number of $(x_1, x_2, x_3, x_4, x_5)$ is $\underline{\hspace{2cm}}$.

Archives

JEE ADVANCED

Single Correct Answer Type

1. The least value of $\alpha \in R$ for which $4\alpha x^2 + \frac{1}{x} \geq 1$, for all $x > 0$, is

- (1) $\frac{1}{64}$ (2) $\frac{1}{32}$
(3) $\frac{1}{27}$ (4) $\frac{1}{25}$

(JEE Advanced 2016)

Numerical Value Type

1. The minimum value of the sum of real numbers a^{-5} , a^{-4} , $3a^{-3}$, 1 , a^8 , and a^{10} with $a > 0$ is _____.

(IIT-JEE 2011)

2. Let m be the minimum possible value of $\log_3(3^{y_1} + 3^{y_2} + 3^{y_3})$ where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^2)$ is _____.

(JEE Advanced 2019)

Answers Key

EXERCISES

Single Correct Answer Type

1. (2) 2. (4) 3. (2) 4. (2) 5. (3)
6. (4) 7. (4) 8. (1) 9. (3) 10. (3)
11. (4) 12. (1) 13. (1) 14. (1) 15. (2)
16. (4) 17. (1) 18. (4) 19. (1) 20. (3)

Multiple Correct Answers Type

1. (1), (2) 2. (1), (3) 3. (1), (2)
4. (3), (4) 5. (1), (2), (3), (4)

Linked Comprehension Type

1. (3) 2. (1) 3. (2) 4. (4) 5. (3)
6. (4)

Numerical Value Type

1. (2) 2. (3) 3. (8) 4. (3) 5. (96)
6. (4) 7. (2) 8. (55) 9. (1)

ARCHIVES

JEE ADVANCED

Single Correct Answer Type

1. (3)

Numerical Value Type

1. (8) 2. (8)

Chapter 6

Concept Application Exercises

Exercise 6.1

$$1. \quad \frac{1}{b} + \frac{1}{c} \geq 2 \left[\frac{1}{b} \frac{1}{c} \right]^{1/2}$$

$$\frac{1}{a} + \frac{1}{b} \geq 2 \left[\frac{1}{a} \frac{1}{b} \right]^{1/2}$$

$$\frac{1}{a} + \frac{1}{c} \geq 2 \left[\frac{1}{a} \frac{1}{c} \right]^{1/2}$$

Adding, we get

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{1}{\sqrt{bc}} + \frac{1}{\sqrt{ca}} + \frac{1}{\sqrt{ab}}$$

2. Put $a = y + z$, $b = z + x$, $c = x + y$, so that $x + y + z = 1$.

As $x = 1 - a$, $y = 1 - b$, $z = 1 - c$ and $0 < a < 1$, $0 < b < 1$, $0 < c < 1$ it follows that $x, y, z > 0$. Now,

$$\frac{a}{1-a} \frac{b}{1-b} \frac{c}{1-c} = \frac{[(y+z)(z+x)(x+y)]}{xyz} \quad (1)$$

Now, A.M. $\geq$ G.M.

$$\Rightarrow \frac{y+z}{2} \geq \sqrt{yx}, \frac{z+x}{2} \geq \sqrt{zx}, \frac{x+y}{2} \geq \sqrt{xy}$$

Multiplying, we get

$$\frac{[(y+z)(z+x)(x+y)]}{(xyz)} \geq 8 \quad (2)$$

From (1) and (2), we get

$$\frac{a}{1-a} \frac{b}{1-b} \frac{c}{1-c} \geq 8$$

3. Using A.M. $\geq$ G.M., we have

$$\frac{1+x^{2n}}{2} \geq (1 \times x^{2n})^{1/2}$$

$$\text{or } \frac{1+x^{2n}}{2} \geq x^n \quad (1)$$

$$\text{and } \frac{1+y^{2m}}{2} \geq (1 \times y^{2m})^{1/2}$$

$$\text{or } \frac{1+y^{2m}}{2} \geq y^m \quad (2)$$

Multiplying (1) and (2), we get

$$\frac{(1+x^{2n})(1+y^{2m})}{4} \geq x^n y^m$$

$$\Rightarrow \frac{x^n y^m}{(1+x^{2n})(1+y^{2m})} \leq \frac{1}{4}$$

$$4. \quad \sqrt{a_1 a_2} < \frac{a_1 + a_2}{2}, \quad (\because \text{G.M.} < \text{A.M.})$$

$$\sqrt{a_1 a_3} < \frac{a_1 + a_3}{2}$$

$\vdots$

$$\sqrt{a_{n-1} a_n} < \frac{a_{n-1} + a_n}{2}$$

Adding, we get

$$\sqrt{a_1 a_2} + \sqrt{a_1 a_3} + \dots < \frac{1}{2} (n-1) (a_1 + a_2 + \dots + a_n)$$

since in all these inequalities in R.H.S. $a_1, a_2, \dots, a_n$ all occur $n-1$ times each.

5. Using A.M. $\geq$ G.M., we have

$$2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{\sin x} 2^{\cos x}}$$

$$= 2\sqrt{2^{\sin x + \cos x}}$$

Now we know that

$$\sin x + \cos x \geq -\sqrt{2}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{-\sqrt{2}}}$$

Hence, the minimum value of $2^{\sin x} + 2^{\cos x}$ is $2^{1-\frac{1}{\sqrt{2}}}$.

6. $\log_{10}(x^3 + y^3) - \log_{10}(x^2 + y^2 - xy) \leq 2$

$$\therefore \log_{10}(x+y)(x^2 + y^2 - xy) - \log_{10}(x^2 + y^2 - xy) \leq 2$$

$$\therefore \log_{10}(x+y) \leq 2$$

$$\therefore x+y \leq 100$$

$$\text{Now } \frac{x+y}{2} \geq (xy)^{1/2}$$

$$\text{or } xy \leq \left(\frac{x+y}{2} \right)^2 \leq \left(\frac{100}{2} \right)^2$$

$$\therefore xy \leq 2500$$

$$\therefore xy_{\max} = 2500$$

7. Using A.M. $\geq$ G.M., we get

$$\frac{a^2 x^4 + b^2 y^4}{2} \geq (a^2 x^4 b^2 y^4)^{1/2}$$

$$\therefore \frac{c^6}{2} \geq (a^2 x^4 b^2 y^4)^{1/2}$$

$$\Rightarrow \frac{c^6}{2} \geq abx^2 y^2$$

$$\Rightarrow xy \leq \frac{c^3}{\sqrt{2ab}}$$

$$\text{Hence, } (xy)_{\max} = \frac{c^3}{\sqrt{2ab}}$$

$$8. \quad \left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) = \frac{1+x+y+xy}{xy} = \frac{9+xy}{xy} = \frac{9}{xy} + 1.$$

Now, using A.M. $\geq$ G.M., we have

$$\frac{x+y}{2} \geq \sqrt{xy}$$

$$\Rightarrow 4 \geq \sqrt{xy}$$

$$\Rightarrow 16 \geq xy$$

$$\Rightarrow \frac{1}{xy} \geq \frac{1}{16}$$

$$\Rightarrow \frac{9}{xy} + 1 \geq \frac{25}{16}$$

Therefore, the minimum value of $\left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right)$ is $\frac{25}{16}$.

Exercise 6.2

1. We have $\frac{1+2+3+4+\dots+n}{n} > (1 \cdot 2 \cdot 3 \cdot 4 \dots n)^{1/n}$

$$\therefore \frac{\frac{n(n+1)}{2}}{n} > (n!)^{1/n}$$

$$\Rightarrow \left(\frac{n+1}{2}\right)^n > n!$$

2. Since A.M. > G.M., we have

$$\frac{1}{n} \left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \frac{a_3}{a_4} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} \right) > \left(\frac{a_1}{a_2} \cdot \frac{a_2}{a_3} \cdot \frac{a_3}{a_4} \dots \frac{a_{n-1}}{a_n} \cdot \frac{a_n}{a_1} \right)^{1/n}$$

$$\text{or } \left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \frac{a_3}{a_4} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} \right) > n(1)^n$$

$$\text{or } \left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \frac{a_3}{a_4} + \dots + \frac{a_{n-1}}{a_n} \right) > n$$

3. In $\triangle ABC$,

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Also,

$$\frac{\tan A + \tan B + \tan C}{3} \geq \sqrt[3]{\tan A \tan B \tan C}$$

[$\because$ A.M. $\geq$ G.M.]

$$\text{or } \tan A \tan B \tan C \geq \sqrt[3]{\tan A \tan B \tan C}$$

$$\text{or } \tan^2 A \tan^2 B \tan^2 C \geq 27 \quad [\text{cubing both sides}]$$

$$\text{or } \tan A \tan B \tan C \geq 3\sqrt{3}$$

$$\text{or } \tan A + \tan B + \tan C \geq 3\sqrt{3}$$

4. We know that,

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1})$$

$$\therefore 3^n - 2^n = 3^{n-1} + 3^{n-2} \cdot 2 + 3^{n-3} \cdot 2^2 + \dots + 2^{n-1}$$

Using A.M $\geq$ G.M., we get

$$\frac{3^{n-1} + 3^{n-2} \cdot 2 + \dots + 2^{n-1}}{n} \geq \left[(3 \cdot 3^2 \cdot \dots \cdot 3^{n-1})(2 \cdot 2^2 \cdot \dots \cdot 2^{n-1}) \right]^{1/n}$$

$$= 3^{\frac{n-1}{2}} \cdot 2^{\frac{n-1}{2}} = 6^{\frac{n-1}{2}}$$

$$\Rightarrow 3^n \geq 2^n + n \cdot 6^{\frac{n-1}{2}}$$

5. Given, $abcd = 1$

We know that, $AM \geq GM$

$$\begin{aligned} \therefore \frac{a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd}{10} &\geq [a^2 \times b^2 \times c^2 \times d^2 \times (ab) \times (ac) \times (ad) \times (bc) \times (bd) \times (cd)]^{1/10} \\ &= (a^5 \times b^5 \times c^5 \times d^5)^{1/10} \\ &= (abcd)^{1/2} \\ &= (1)^{1/2} \quad [\because abcd = 1] \end{aligned}$$

$$\Rightarrow \frac{a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd}{10} \geq 1$$

$$\Rightarrow a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd \geq 10$$

6. Using A.M. $\geq$ H.M., we have

$$\frac{(x+y) + (y+z) + (z+x)}{3} \geq \frac{3}{\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x}}$$

$$\Rightarrow 2(x+y+z) \left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x} \right) \geq 9$$

$$\Rightarrow 2 \left(\frac{x+y+z}{x+y} + \frac{x+y+z}{y+z} + \frac{x+y+z}{z+x} \right) \geq 9$$

$$\Rightarrow 2 \left[\left(1 + \frac{z}{x+y} \right) + \left(1 + \frac{x}{y+z} \right) + \left(1 + \frac{y}{z+x} \right) \right] \geq 9$$

$$\Rightarrow 2 \left(\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y} \right) \geq 3$$

$$\Rightarrow 2 \left(\frac{x}{1-x} + \frac{y}{1-y} + \frac{z}{1-z} \right) \geq 3 \quad (\because x+y+z=1)$$

Exercise 6.3

1. $\frac{[a + a + \dots a \text{ times}] + [b + b + \dots b \text{ times}]}{a+b} \geq [a^a b^b]^{\frac{1}{a+b}}$

$$\geq \frac{a+b}{\left(\frac{1}{a} + \frac{1}{a} + \dots a \text{ times} \right) + \left(\frac{1}{b} + \frac{1}{b} + \dots b \text{ times} \right)}$$

$$\text{or } \frac{a^2 + b^2}{a+b} \geq [a^a b^b]^{\frac{1}{a+b}} \geq \frac{a+b}{1+1}$$

$$\text{or } \left(\frac{a^2 + b^2}{a+b} \right)^{a+b} \geq a^a b^b \geq \left(\frac{a+b}{2} \right)^{a+b}$$

$$\text{or } \left[\frac{a^2 + b^2}{a+b} \right]^{a+b} > a^a b^b > \left\{ \frac{a+b}{2} \right\}^{a+b}$$

2. Consider p quantities each equal to a and q quantities each equal to b . We know that

A.M. > G.M.

$$\therefore \frac{(a + a + \dots \text{to } p \text{ terms}) + (b + b + \dots \text{to } q \text{ terms})}{p+q}$$

$$> [(a \times a \dots \text{to } p \text{ factors})(b \times b \dots \text{to } q \text{ factors})]^{\frac{1}{p+q}}$$

$$\Rightarrow \frac{ap + bq}{p+q} > (a^p b^q)^{\frac{1}{p+q}}$$

$$\Rightarrow \left(\frac{ap + bq}{p+q} \right)^{p+q} > (a^p b^q)$$

3. We know that

Weighted A.M. > weighed G.M.

$$\Rightarrow \frac{px^{q-r} + qx^{r-p} + rx^{p-q}}{p+q+r} > \left[(x^{q-r})^p (x^{r-p})^q (x^{p-q})^r \right]^{\frac{1}{p+q+r}}$$

$$\text{or } \frac{px^{q-r} + qx^{r-p} + rx^{p-q}}{p+q+r} > 1$$

$$\text{or } px^{q-r} + qx^{r-p} + rx^{p-q} > p + q + r$$

In case either $p = q = r$ or $x = 1$, inequality becomes equality.

4. Using weighted A.M., G.M. inequality, we have

$$\frac{ca + ab + bc}{a + b + c} \geq (a^c b^a c^b)^{\frac{1}{a+b+c}}$$

$$\text{or } ca + ab + bc \geq (a^c b^a c^b)$$

Similarly,

$$\frac{ba + cb + ac}{b + c + a} \geq (a^b b^c c^a)^{\frac{1}{a+b+c}} \text{ or } ba + cb + ac \geq (a^b b^c c^a)$$

$$\frac{aa + bb + cc}{a + b + c} \geq (a^a b^b c^c)^{\frac{1}{a+b+c}} \text{ or } aa + bb + cc \geq (a^a b^b c^c)$$

Adding together, we get

$$(a + b + c)(a + b + c) \geq a^a b^b c^c + a^b b^c c^a + a^c b^a c^b$$

$$\Rightarrow 1 \geq a^a b^b c^c + a^b b^c c^a + a^c b^a c^b$$

5. Using A.M. $\geq$ G.M., we get

$$\frac{\frac{a^2}{2} + \frac{a^2}{2} + b^2}{3} \geq \left(\frac{a^2}{2} \cdot \frac{a^2}{2} \cdot b^2 \right)^{\frac{1}{3}}$$

$$\Rightarrow \frac{4}{3} \geq \frac{(a^2 b)^{\frac{2}{3}}}{2^{\frac{2}{3}}}$$

$$\Rightarrow (a^2 b) \leq \left(\frac{4}{3} \right)^{\frac{3}{2}} \times 2 = \frac{16}{3\sqrt{3}}$$

Therefore, maximum value of $a^2 b$ is $\frac{16}{3\sqrt{3}}$.

6. We have

$$\frac{2\frac{x^2}{2} + 3\frac{y^2}{3} + 4\frac{z^2}{4}}{9} \geq \left(\left(\frac{x^2}{2} \right)^2 \left(\frac{y^2}{3} \right)^3 \left(\frac{z^2}{4} \right)^4 \right)^{\frac{1}{9}}$$

$$\Rightarrow \frac{1}{9^9} \geq \frac{x^4 y^6 z^8}{2^2 3^3 4^4}$$

$$\Rightarrow x^4 y^6 z^8 \leq \frac{2^2 3^3 4^4}{9^9}$$

$$\Rightarrow x^2 y^3 z^4 \leq 2^5 \times 3^{-\frac{15}{2}}$$

Exercise 6.4

$$1. \frac{a^4 + b^4 + c^4}{3} > \left(\frac{a + b + c}{3} \right)^4$$

$$= \left(\frac{a + b + c}{3} \right) \left(\frac{a + b + c}{3} \right)^3$$

Now,

$$\frac{a + b + c}{3} > (abc)^{1/3} \text{ or } \left(\frac{a + b + c}{3} \right)^3 > abc$$

$$\Rightarrow \frac{a^4 + b^4 + c^4}{3} > \left(\frac{a + b + c}{3} \right) abc$$

$$\text{or } a^4 + b^4 + c^4 > abc(a + b + c)$$

2. A.M. of $(1/2)$ th powers $<$ $(1/2)$ th power of A.M.

$$\therefore \frac{(C_1)^{\frac{1}{2}} + (C_2)^{\frac{1}{2}} + \dots + (C_n)^{\frac{1}{2}}}{n} < \left(\frac{C_1 + C_2 + \dots + C_n}{n} \right)^{1/2}$$

$$\text{or } \frac{\sqrt{C_1} + \sqrt{C_2} + \dots + \sqrt{C_n}}{n} < \left(\frac{2^n - 1}{n} \right)^{1/2}$$

$$\text{or } \sqrt{C_1} + \sqrt{C_2} + \dots + \sqrt{C_n} < \frac{n\sqrt{(2^n - 1)}}{\sqrt{n}}$$

Hence,

$$\sqrt{C_1} + \sqrt{C_2} + \dots + \sqrt{C_n} < \sqrt{[n(2^n - 1)]}$$

3. We know that A.M. of m th power $>$ m th power of A.M.

$$\therefore \frac{\left(a + \frac{1}{a} \right)^2 + \left(b + \frac{1}{b} \right)^2}{2} > \left[\frac{\left(a + \frac{1}{a} \right) + \left(b + \frac{1}{b} \right)}{2} \right]^2, \text{ here } m = 2$$

$$\text{or } \left(a + \frac{1}{a} \right)^2 + \left(b + \frac{1}{b} \right)^2 > \frac{1}{2} \left[(a + b) + \left(\frac{1}{a} + \frac{1}{b} \right) \right]^2 \quad (1)$$

$$\text{Also, } \frac{a^{-1} + b^{-1}}{2} > \left(\frac{a + b}{2} \right)^{-1}$$

$$\text{or } \frac{1}{2} \left(\frac{1}{a} + \frac{1}{b} \right) > \frac{2}{a + b}$$

$$\text{or } \frac{1}{a} + \frac{1}{b} > \frac{4}{a + b}$$

$$\text{or } \frac{1}{a} + \frac{1}{b} > 4$$

Hence, from (1),

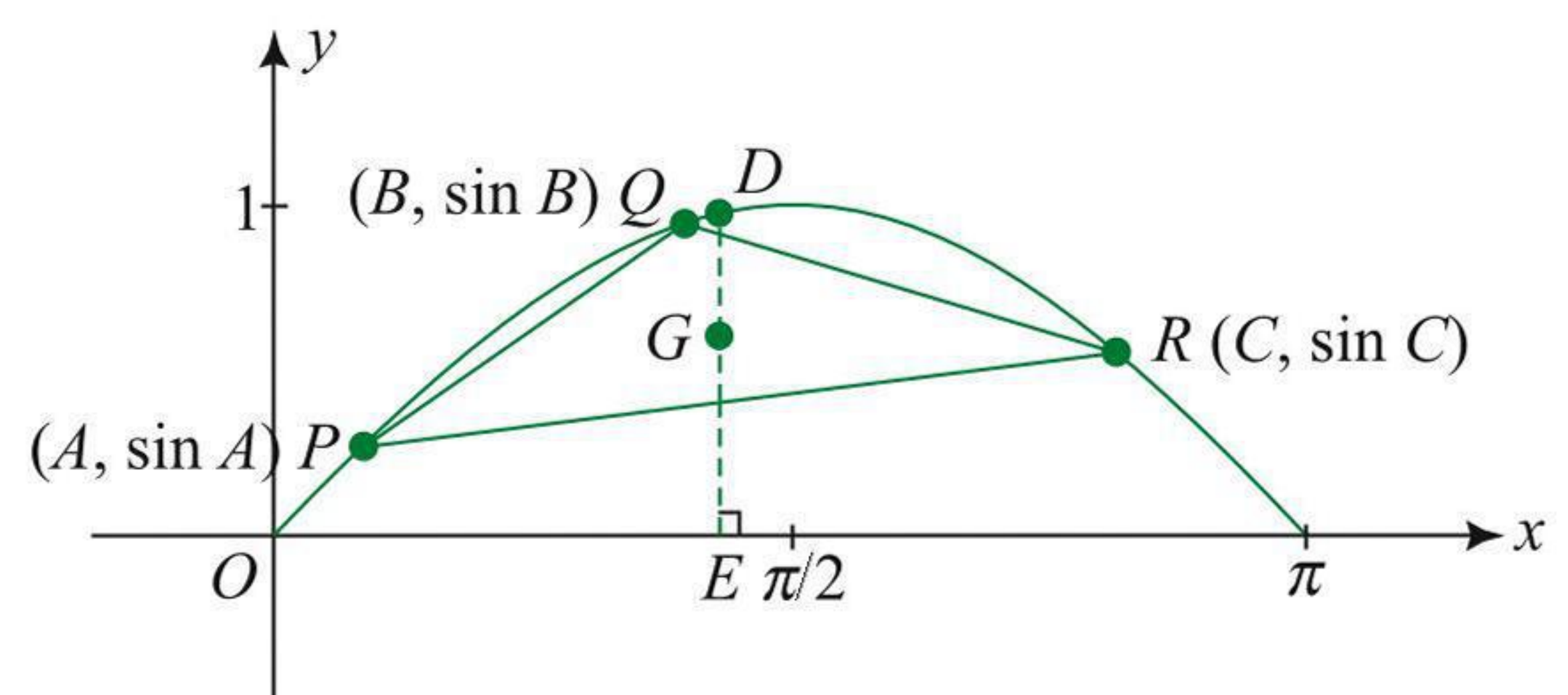
$$\text{From (i), } \left(a + \frac{1}{a} \right)^2 + \left(b + \frac{1}{b} \right)^2 > \frac{1}{2} (1 + 4)^2 = \frac{25}{2}$$

4. In $\triangle ABC$, we have $0 < A, B, C < \pi$.

So, consider the graph of function $y = \sin x$ for $x \in (0, \pi)$.

The graph of function is concave downward.

Now, consider three points on the graph $P(A, \sin A)$, $Q(B, \sin B)$ and $R(C, \sin C)$.



Join these points to get triangle PQR .

Now, centroid of the triangle PQR is

$$G \left(\frac{A + B + C}{3}, \frac{\sin A + \sin B + \sin C}{3} \right).$$

Through point G , draw a line parallel to y -axis meeting graph at point D and x -axis at point E .

The abscissa of point D is $\frac{A+B+C}{3}$.

So, coordinates of point D are $\left(\frac{A+B+C}{3}, \sin\left(\frac{A+B+C}{3}\right)\right)$.

Clearly, $DE \geq GE$

$$\therefore \sin\left(\frac{A+B+C}{3}\right) \geq \frac{\sin A + \sin B + \sin C}{3}$$

$$\Rightarrow \sin \frac{\pi}{3} \geq \frac{\sin A + \sin B + \sin C}{3}$$

$$\Rightarrow \sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

Exercises

Single Correct Answer Type

1. (2) Using A.M. $\geq$ G.M., we have

$$x^4 + y^4 \geq 2x^2y^2 \text{ and } 2x^2y^2 + z^2 \geq \sqrt{8}xyz$$

$$\Rightarrow \frac{x^4 + y^4 + z^2}{xyz} \geq \sqrt{8}$$

2. (4) $a^2 + b^2 = k^2$

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 = a^2 + b^2 + 4 + \frac{1}{a^2} + \frac{1}{b^2}$$

$$= k^2 + 4 + \frac{k^2}{a^2b^2}$$

$$\geq k^2 + 4 + \frac{k^2}{\left(\frac{a^2+b^2}{2}\right)^2}$$

$$= k^2 + 4 + \frac{4}{k^2}$$

3. (2) Applying A.M. $\geq$ G.M. in $6 \tan^2 \phi$, $54 \cot^2 \phi$, 18, we get

$$\frac{6 \tan^2 \phi + 54 \cot^2 \phi + 18}{3} \geq (6 \times 54 \times 18)^{1/3} \geq 18$$

Now equality holds when

$$6 \tan^2 \phi = 54 \cot^2 \phi = 18$$

$$\Rightarrow \tan^2 \phi = 3 \text{ and } \cot^2 \phi = \frac{1}{3}$$

Hence, the statements I and II are correct.

4. (2) A.M. $\geq$ G.M.

$$\Rightarrow \frac{a+b+c}{3} \geq (abc)^{1/3}$$

$$\text{or } a+b+c \geq 3(abc)^{1/3} \quad (1)$$

But given ab^2c^3 , $a^2b^3c^4$, $a^3b^4c^5$ are in A.P. ($\because abc \neq 0$). Hence,

$$2abc = 1 + a^2b^2c^2$$

$$\Rightarrow (abc - 1)^2 = 0$$

$$\therefore abc = 1$$

Now from Eq. (1), we get

$$a+b+c \geq 3(1)^{1/3}$$

$$\text{or } (a+b+c) \geq 3$$

Hence, minimum value of $a+b+c$ is 3.

$$5. (3) y = \frac{3^x}{3} + \frac{1}{3^x 3}$$

Now,

$$\text{A.M.} \geq \text{G.M.}$$

$$\Rightarrow \frac{\frac{3^x}{3} + \frac{1}{3^x 3}}{2} \geq \sqrt{\frac{3^x}{3} \cdot \frac{1}{3^x 3}}$$

$$\text{or } 3^{x-1} + 3^{-x-1} \geq \frac{2}{3}$$

6. (4) We have A.M. $\geq$ G.M. Therefore,

$$\frac{a}{b} + \frac{b}{a} + \frac{b}{c} + \frac{c}{b} + \frac{c}{a} + \frac{a}{c} \geq 6$$

$$\text{or } \frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} \geq 6$$

Hence, the least value is 6.

7. (4) Let $a_1, a_2, \dots, a_n$ be n positive numbers such that $a_1 a_2 \dots a_n = n^n$. Since A.M. $\geq$ G.M., we have

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}$$

$$\text{or } \frac{a_1 + a_2 + \dots + a_n}{n} \geq n$$

$$\text{or } a_1 + a_2 + \dots + a_n \geq n^2$$

8. (1) A.M. $\geq$ G.M.

$$\Rightarrow \frac{bcx + cay + abz}{3} \geq (a^2 b^2 c^2 xyz)^{1/3}$$

$$\text{or } bcx + cay + abz \geq 3xyz$$

$$\text{or } bcx + acy + abz \geq 3abc$$

9. (3) Consider $1/l$, $2/m$, $3/n$ and use A.M. $\geq$ G.M. Then,

$$\frac{1}{3} \left(\frac{1}{l} + \frac{2}{m} + \frac{3}{n} \right) \geq \left(\frac{1}{l} \cdot \frac{2}{m} \cdot \frac{3}{n} \right)^{1/3} = \left(\frac{6}{lmn} \right)^{1/3}$$

But $lmn = 48$.

$$\therefore \frac{1}{3} \left(\frac{1}{l} + \frac{2}{m} + \frac{3}{n} \right) \geq \left(\frac{6}{48} \right)^{1/3} = \frac{1}{2}$$

$$\therefore \left(\frac{1}{l} + \frac{2}{m} + \frac{3}{n} \right)_{\min} = \frac{3}{2}$$

10. (3) a, b, c are in H.P. Hence, H.M. of a and c is b .

$$\therefore \sqrt{ac} > b \quad (\because \text{G.M.} > \text{H.M.})$$

Since A.M. $>$ G.M., so

$$\frac{a^{101} + c^{101}}{2} > (\sqrt{ac})^{101} > b^{101} \quad (\because \sqrt{ac} > b)$$

$$\text{or } 2b^{101} - a^{101} - c^{101} < 0$$

$$\text{Let } f(x) = x^2 - kx + 2b^{101} - a^{101} - c^{101}$$

$$\therefore f(0) = 2b^{101} - a^{101} - c^{101} < 0$$

Hence, equation $f(x) = 0$ has one root in $(-\infty, 0)$ and other in $(0, \infty)$.

$$\begin{aligned}
 11. (4) \quad a &= \frac{x^2 + 4}{|x|} - 3 \\
 &= |x| + \frac{4}{|x|} - 3 \\
 &\geq 2\sqrt{|x| \times \frac{4}{|x|}} - 3 \quad (\because \text{A.M.} \geq \text{G.M.}) \\
 &\Rightarrow a \geq 1
 \end{aligned}$$

$$\begin{aligned}
 12. (1) \quad 2E &= \frac{2a}{b+c-a} + \frac{2b}{c+a-b} + \frac{2c}{a+b-c} \\
 &= \frac{2a}{b+c-a} + 1 + \frac{2b}{c+a-b} + 1 + \frac{2c}{a+b-c} + 1 - 3 \\
 &= (a+b+c) \left(\frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} \right) - 3
 \end{aligned}$$

Using A.M. $\geq$ H.M. we have

$$\begin{aligned}
 \frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} &\geq \frac{3}{a+b+c} \\
 \text{or } (a+b+c) \left(\frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} \right) &\geq 9 \\
 \text{or } (a+b+c) \left(\frac{1}{b+c-a} + \frac{1}{c+a-b} + \frac{1}{a+b-c} \right) - 3 &\geq 6 \\
 \Rightarrow E &\geq 3
 \end{aligned}$$

13. (1) A.M. $\geq$ G.M.

$$\begin{aligned}
 \Rightarrow \frac{\log_d a + \log_c b + \log_a c + \log_b d}{4} \\
 \geq \sqrt[4]{\frac{\log a}{\log d} \times \frac{\log b}{\log c} \times \frac{\log c}{\log a} \times \frac{\log d}{\log b}}
 \end{aligned}$$

$$\text{or } \log_d a + \log_b d + \log_a c + \log_c b \geq 4$$

14. (1) Using A.M. and H.M. inequality, we get

$$\frac{2bc}{b+c} \leq \frac{b+c}{2}, \quad \frac{2ac}{a+c} \leq \frac{a+c}{2}, \quad \frac{2ab}{a+b} \leq \frac{a+b}{2}$$

$$\text{Adding, } \frac{bc}{b+c} + \frac{ac}{a+c} + \frac{ab}{a+b} \leq \frac{1}{2}(a+b+c)$$

$$15. (2) \quad \frac{a+b+c}{3} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \quad (\because \text{A.M.} \geq \text{H.M.})$$

$$\text{or } (a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$$

$$16. (4) \quad a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) = ab^2 + ac^2 + bc^2 + ba^2 + ca^2 + cb^2$$

Using A.M. $\geq$ G.M., we get

$$\frac{ab^2 + ac^2 + bc^2 + ba^2 + ca^2 + cb^2}{6} \geq (a^6 b^6 c^6)^{1/6}$$

$$\text{or } a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) \geq 6abc$$

17. (1) a, b, c, d are in H.P. Hence, b is H.M. of a and c ; c is H.M. of b and d . Using A.M. $\geq$ H.M., we get

$$\frac{a+c}{2} > b \quad \text{or } a+c > 2b$$

$$\begin{aligned}
 &\text{and } \frac{b+d}{2} > c \quad \text{or } b+d > 2c \\
 &\text{or } a+c+b+d > 2b+2c \\
 &\text{or } a+d > b+c
 \end{aligned}$$

18. (4) $a+b+c=18$

$$\Rightarrow 2 \times \frac{a}{2} + 3 \times \frac{b}{3} + 4 \times \frac{c}{4} = 18$$

Using weighted A.M. and G.M. inequality, we get

$$\frac{2 \times \frac{a}{2} + 3 \times \frac{b}{3} + 4 \times \frac{c}{4}}{9} \geq \left(\left(\frac{a}{2} \right)^2 \left(\frac{b}{3} \right)^3 \left(\frac{c}{4} \right)^4 \right)^{1/9}$$

$$\text{or } 2^9 \geq \frac{a^2}{2^2} \times \frac{b^3}{3^3} \times \frac{c^4}{4^4}$$

$$\text{or } a^2 b^3 c^4 \leq 3^3 \times 2^{19}$$

19. (1) Let

$$\begin{aligned}
 x-3 &= t \\
 \Rightarrow x-2 &= (t+1) \quad \text{and } x-1 = t+2 \\
 \Rightarrow f(x) &= \frac{(x-2)(x-1)}{(x-3)} \\
 &= \frac{(t+1)(t+2)}{t} \\
 &= \frac{(t^2 + 3t + 2)}{t} \\
 &= t + \frac{2}{t} + 3 \\
 &\geq 3 + 2\sqrt{2} \quad (\text{using A.M.} \geq \text{G.M., as } t > 0)
 \end{aligned}$$

20. (3) We have

$$\frac{a^3 + 1}{2} \geq \sqrt{a^3 \times 1}$$

$$\text{and } \frac{a^2 + a^1}{2} \geq \sqrt{a^2 a^1}$$

Adding, we get

$$\frac{a^3 + a^2 + a + 1}{2} \geq 2\sqrt{a^3}$$

Multiple Correct Answers Type

1. (1), (2)

We have

$$2s = a + b + c$$

$$A^2 = s(s-a)(s-b)(s-c)$$

Now,

$$\text{A.M.} \geq \text{G.M.}$$

$$\Rightarrow \frac{s + (s-a) + (s-b) + (s-c)}{4} \geq [s(s-a)(s-b)(s-c)]^{1/4}$$

$$\Rightarrow \frac{4s - 2s}{4} \geq [A^2]^{1/4}$$

$$\Rightarrow s/2 \geq A^{1/2} \quad \text{or } A \leq s^2/4$$

Also,

$$\frac{(s-a) + (s-b) + (s-c)}{3} \geq [(s-a)(s-b)(s-c)]^{1/3}$$

$$\Rightarrow \frac{s}{3} \geq \left[\frac{A^2}{s} \right]^{1/3}$$

$$\text{or } \frac{A^2}{s} \leq \frac{s^3}{27}$$

$$\text{or } A \leq \frac{s^2}{3\sqrt{3}}$$

2. (1), (3)

A.M. of x and z is y . G.M. of x and z is $\sqrt{xz}$.

Now, A.M. $\geq$ G.M. $\Rightarrow y^2 \geq xz$

$$\text{Also, A.M.} \geq \text{H.M.} \Rightarrow y \geq \frac{2xz}{x+z}$$

$$\frac{x+y}{2y-x} = \frac{x+y}{x+z-x} = \frac{x+y}{z} \quad \text{and} \quad \frac{y+z}{2y-z} = \frac{y+z}{x}$$

$$\begin{aligned} \therefore \frac{\frac{x+y}{2y-x} + \frac{y+z}{2y-z}}{2} &\geq \sqrt{\frac{x+y}{z} \cdot \frac{y+z}{x}} \\ &= \sqrt{1 + \frac{y(x+y+z)}{xz}} \quad [\because x+z=2y] \\ &= \sqrt{1 + \frac{3y^2}{xz}} \end{aligned}$$

$$\therefore \frac{x+y}{2y-x} + \frac{y+z}{2y-z} \geq 2\sqrt{1 + \frac{3y^2}{xz}} \geq 4 \quad [\because y^2 \geq xz]$$

3. (1), (2)

Using A.M. $\geq$ G.M. one can show that

$$(b+c)(c+a)(a+b) \geq 8abc$$

$$\Rightarrow (p-a)(p-b)(p-c) \geq 8abc$$

Therefore, (2) holds. Also,

$$\frac{(p-a) + (p-b) + (p-c)}{3} \geq [(p-a)(p-b)(p-c)]^{1/3}$$

$$\text{or } \frac{3p - (a+b+c)}{3} \geq [(p-a)(p-b)(p-c)]^{1/3}$$

$$\text{or } \frac{2p}{3} \geq [(p-a)(p-b)(p-c)]^{1/3}$$

$$\text{or } (p-a)(p-b)(p-c) \leq \frac{8p^3}{27}$$

Therefore, (1) holds. Again,

$$\frac{1}{2} \left(\frac{bc}{a} + \frac{ca}{b} \right) \geq \sqrt{\left(\frac{bc}{a} \right) \left(\frac{ca}{b} \right)}$$

and so on. Adding the inequalities, we get

$$\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \geq a+b+c = p$$

Therefore, (3) does not hold.

4. (3), (4)

Consider the A.P. Since a is equidistant from the first term α and the last term β of the A.P., therefore, α, a, β are in A.P. Hence, a is the A.M. of α and β . So,

$$a = \frac{\alpha + \beta}{2}$$

Similarly, b and c are the geometric and harmonic means, i.e.,

$$b = \sqrt{\alpha\beta} \quad \text{and} \quad c = \frac{2\alpha\beta}{\alpha + \beta}$$

Since A.M., G.M., and H.M. are in G.P. and A.M. $\geq$ G.M. $\geq$ H.M., therefore, a, b, c are in G.P. and $a \geq b \geq c$.

5. (1), (2), (3), (4)

A.M. $\geq$ G.M.

$$\begin{aligned} \Rightarrow \frac{\frac{2a}{4} + \frac{2a}{4} + \frac{2a}{4} + \frac{2a}{4} + \frac{b}{2} + \frac{b}{2} + \frac{3c}{2} + \frac{3c}{2}}{8} \\ \geq \sqrt[8]{\frac{2a}{4} \cdot \frac{2a}{4} \cdot \frac{2a}{4} \cdot \frac{2a}{4} \cdot \frac{b}{2} \cdot \frac{b}{2} \cdot \frac{3c}{2} \cdot \frac{3c}{2}} \\ \Rightarrow \frac{2a+b+3c}{8} \geq \left(\frac{3^2}{2^8} a^4 b^2 c^2 \right)^{1/8} \end{aligned}$$

The greatest value takes place when A.M. = G.M. and

$$\frac{2a}{4} = \frac{b}{2} = \frac{3c}{2}$$

$$\Rightarrow a = b = 3c = K$$

$$\text{Now, } 2a + b + 3c = 1$$

$$\Rightarrow 2K + K + K = 1$$

$$\Rightarrow K = 1/4$$

$$\Rightarrow a = b = 1/4 \text{ and } c = 1/12$$

Linked Comprehension Type

1. (3), 2. (1), 3. (2)

Let roots of equation $x^6 - 12x^5 + bx^4 + cx^3 + dx^2 + ex + 64 = 0$ be $x_i, i = 1, 2, \dots, 6$. Now,

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 12$$

$$\text{and } x_1 x_2 x_3 x_4 x_5 x_6 = 64$$

Thus,

$$\frac{x_1 + x_2 + \dots + x_6}{6} = 2 \quad \text{and} \quad (x_1 x_2 x_3 x_4 x_5 x_6)^{1/6} = 2$$

$$\Rightarrow \text{A.M.} = \text{G.M.}$$

$$\Rightarrow x_1 = x_2 = x_3 = x_4 = x_5 = x_6 = 2$$

Hence, the given equation is equivalent to

$$(x-2)^6 = 0$$

$$\text{or } x^6 - 12x^5 + 60x^4 - 160x^3 + 240x^2 - 192x - 64 = 0$$

$$\therefore f(1) = 1 - 12 + 60 - 160 + 240 - 192 + 64 = 1$$

4. (4), 5. (3), 6. (4)

Since, $a, b, c \geq 0$ roots must be negative. Let the roots be $x_1, x_2, x_3, x_4 (< 0)$. Then,

$$x_1 + x_2 + x_3 + x_4 = -a$$

$$x_1 x_2 + x_1 x_3 + x_1 x_4 + x_2 x_3 + x_2 x_4 + x_3 x_4 = b$$

$$x_1 x_2 x_3 + x_2 x_3 x_4 + x_3 x_4 x_1 + x_4 x_1 x_2 = -c$$

$$x_1 x_2 x_3 x_4 = 1$$

Now,

$$\frac{(-x_1) + (-x_2) + (-x_3) + (-x_4)}{4} \geq [(-x_1)(-x_2)(-x_3)(-x_4)]^{1/4}$$

$$(\because \text{A.M.} \geq \text{G.M.})$$

$$\Rightarrow \frac{a}{4} \geq 1$$

$$\text{or } a \geq 4$$

Hence, the minimum value of a is 4. Similarly,

$$\frac{x_1 x_2 + x_1 x_3 + \dots + x_3 x_4}{6} \geq [x_1^3 x_2^3 x_3^3 x_4^3]^{1/4}$$

$$\Rightarrow \frac{b}{6} \geq 1$$

$$\text{or } b \geq 6$$

Hence, the minimum value of b is 6. Finally,

$$\frac{-x_1x_2x_3 - x_2x_3x_4 - x_3x_4x_1 - x_4x_1x_2}{4} \geq [x_1^3x_2^3x_3^3x_4^3]^{1/4}$$

$$\Rightarrow \frac{c}{4} \geq 1$$

$$\text{or } c \geq 4$$

Hence, the minimum value of c is 4.

Numerical Value Type

$$1. (2) f(x) = \frac{4x^2 + 8x + 13}{6(1+x)}$$

$$= \frac{4(x+1)^2 + 9}{6(1+x)}$$

$$= \frac{2}{3}(x+1) + \frac{3}{2(x+1)}$$

$$\geq 2\sqrt{\frac{2}{3} \cdot \frac{3}{2}} = 2 \quad (\text{A.M.} \geq \text{G.M.})$$

Therefore, minimum value of $f(x)$ is 2.

$$2. (3) a + b = 3$$

H.M. $\leq$ A.M. for 3 numbers $\frac{a}{2}, \frac{a}{2}, b$. We have

$$\frac{3}{\frac{2}{a} + \frac{2}{a} + \frac{1}{b}} \leq \frac{\frac{a}{2} + \frac{a}{2} + b}{3} = 1$$

$$\therefore 1 \geq \frac{3}{\frac{2}{a} + \frac{2}{a} + \frac{1}{b}} \text{ or } \frac{2}{a} + \frac{2}{a} + \frac{1}{b} \geq 3$$

$$\therefore \frac{4}{a} + \frac{1}{b} \geq 3$$

$$3. (8) \text{ Consider values } xy, yz, zx$$

Now A.M. $\geq$ G.M.

$$\Rightarrow \frac{xy + yz + zx}{3} \geq (x^2y^2z^2)^{1/3}$$

$$\Rightarrow 4^{3/2} \geq xyz$$

$$\text{or } xyz \leq 8$$

$$4. (3) 9a + 3b + c = 90$$

$$\Rightarrow 3a + b + \frac{c}{3} = 30$$

Now consider numbers $3a, b$, and $\frac{c}{3}$

$$\left(3a \times b \times \frac{c}{3}\right)^{1/3} \leq \frac{3a + b + \frac{c}{3}}{3} \quad (\text{as GM} \leq \text{AM})$$

$$\text{or } (abc)^{1/3} \leq \frac{30}{3} = 10$$

$$\text{or } abc \leq 1000$$

$$\Rightarrow \log a + \log b + \log c \leq 3$$

$$5. (96) \text{ Using AM} \geq \text{GM for } x^2, 2xy, 2xy, 4y^2, z^2, z^2$$

$$\therefore \frac{x^2 + 2xy + 2xy + 4y^2 + z^2 + z^2}{6} \geq [16(xyz)^4]^{1/6}$$

$$= [16(32)^4]^{1/6} = (2^{24})^{1/6} = 16$$

$$6. (4) \text{ We have } \frac{2\left(\frac{x}{2}\right) + y}{3} \geq \left(\left(\frac{x}{2}\right)^2 y\right)^{1/3}$$

$$\Rightarrow \left(\frac{3}{2}\right)^3 \geq \frac{x^2 y}{4}$$

$$\Rightarrow x^2 y \leq 4$$

Therefore, maximum value of $x^2 y$ is 4.

$$7. (2) \text{ As } x, y \in R \text{ and } xy > 0, \text{ so } x \text{ and } y \text{ will be of same sign.}$$

Therefore, all the quantities $\frac{2x}{y^3}, \frac{x^3 y}{3}, \frac{4y^2}{9x^4}$ are positive.

Now A.M. $\geq$ G.M.

$$\Rightarrow \frac{2x}{y^3} + \frac{x^3 y}{3} + \frac{4y^2}{9x^4} \geq 3 \left(\left(\frac{2x}{y^3} \right) \left(\frac{x^3 y}{3} \right) \left(\frac{4y^2}{9x^4} \right) \right)^{1/3}$$

$$= 3 \times \frac{2}{3} = 2$$

$$8. (55) \left(a + \frac{b}{2} + \frac{b}{2} + \frac{c}{3} + \frac{c}{3} + \frac{c}{3} + \frac{d}{4} + \frac{d}{4} + \frac{d}{4} + \frac{d}{4} + \frac{e}{5} + \frac{e}{5} + \frac{e}{5} + \frac{e}{5} \right) / 15 = 1$$

$$\therefore \text{A.M.} = 1$$

$$\text{G.M.} = \left(\frac{ab^2c^3d^4e^5}{1 \cdot 2^2 \cdot 3^3 \cdot 4^4 \cdot 5^5} \right)^{1/15}$$

$$= \left(\frac{(120)^3 \times 50}{(3 \times 5 \times 8)^3 \times 2 \times 5^2} \right)^{1/15} = 1$$

$$\therefore \text{A.M.} = \text{G.M.}$$

$$\Rightarrow a = \frac{b}{2} = \frac{c}{3} = \frac{d}{4} = \frac{e}{5} = \lambda$$

$$\Rightarrow \lambda = 1$$

$$\therefore a = 1, b = 2, c = 3, d = 4, e = 5$$

$$\therefore a^2 + b^2 + c^2 + d^2 + e^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

$$9. (1) \text{ Using A.M.} \geq \text{G.M., we get}$$

$$x_2^2 + 1 \geq 2x_2$$

$$x_3^3 + 1 + 1 \geq 3x_3$$

$$x_4^4 + 1 + 1 + 1 \geq 4x_4$$

$$x_5^5 + 1 + 1 + 1 + 1 \geq 5x_5$$

Adding these, we get

$$x_1 + x_2^2 + x_3^3 + x_4^4 + x_5^5 + 10 \geq x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5$$

$$x_1 + x_2^2 + x_3^3 + x_4^4 + x_5^5 = 5$$

The equality holds if $x_1 = x_2 = x_3 = x_4 = x_5 = 1$.

So, there is only one possibility for $(x_1, x_2, x_3, x_4, x_5)$.

Archives

JEE ADVANCED

Single Correct Answer Type

1. (3) Using A.M. $\geq$ G.M.

$$\frac{4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x}}{3} \geq \sqrt[3]{4\alpha x^2 \cdot \frac{1}{4x^2}}$$

$$\Rightarrow 4\alpha x^2 + \frac{1}{x} \geq 3\sqrt[3]{\alpha}$$

If this is true for all $x > 0$, then

$$3\sqrt[3]{\alpha} \geq 1 \Rightarrow \alpha \geq \frac{1}{27}$$

Numerical Value Type

1. (8) Using A.M. $\geq$ G.M.

$$\Rightarrow \frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + a^8 + a^{10} + 1}{8} \geq 1$$

$$\text{or } a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + a^8 + a^{10} + 1 \geq 8$$

$$\text{or } (a^{-5} + a^{-4} + 3a^{-3} + a^8 + a^{10} + 1)_{\min} = 8$$

2. (8)

Using A.M. $\geq$ G.M., we have

$$\frac{3^{y_1} + 3^{y_2} + 3^{y_3}}{3} \geq [3^{(y_1 + y_2 + y_3)}]^{1/3}$$

$$\Rightarrow 3^{y_1} + 3^{y_2} + 3^{y_3} \geq 3^4$$

$$\Rightarrow \log_3(3^{y_1} + 3^{y_2} + 3^{y_3}) \geq 4$$

$$\Rightarrow m = 4$$

$$\text{Also, } \frac{x_1 + x_2 + x_3}{3} \geq \sqrt[3]{x_1 x_2 x_3}$$

$$\Rightarrow x_1 x_2 x_3 \leq 27$$

$$\Rightarrow (\log_3 x_1 + \log_3 x_2 + \log_3 x_3) \leq 3$$

$$\Rightarrow M = 3$$

$$\text{Thus, } \log_2(m^3) + \log_3(M^2) = \log_2(4^3) + \log_3(3^2) \\ = 6 + 2 = 8$$